An open curtailment dataset and an evaluation of adaptive conformal calibration under regime change

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Abstract

Curtailment of variable renewable generation is zero-inflated, heavy-tailed and drifts as the system changes, a demanding target for calibrated uncertainty. We study it in Chile's National Electricity System, where about 19,000 GWh were curtailed between 2022 and mid-2026. We release an open, errata-corrected, plant-level dataset of daily and hourly curtailment (53 months, 300 plants) that reconciles exactly with the operator's annual totals, and use it to evaluate conformal calibration of seven-day-ahead upper limits under a documented distributional change. With a lognormal hurdle base model, an adaptive scheme combining score transport and online recalibration narrows the limits of the most divergent window by a third at near-nominal coverage. Across nine base models from two families the gain does not transfer, and it is largely mechanical: a constant-threshold oracle that matches the adaptive coverage without adapting in time explains 84% of the variation in width change across 45 model-window cells, as a controlled simulation confirms. The gain measures the drift of the base model's calibration, which here depends more on the base model than on the window. Improving the base model dominates: early stopping alone lowers the hurdle's interval score by 35%. On high-curtailment days defined by the outcome every method loses coverage, and the transport-based scheme, which leaves more curtailed energy uncovered than a static split, is not the method we recommend for operation. A static conformal wrapper around a multi-quantile predictive gives the lowest operational cost among methods with near-nominal coverage. Dataset and code are public.

Keywords: renewable curtailment, probabilistic forecasting, conformal prediction, distribution shift, forecast evaluation, power system operations

1. Introduction

The share of variable renewable energy has grown faster than the transmission and demand able to absorb it, and curtailment, the deliberate reduction of available solar, wind or hydropower output, has become a routine consequence of that imbalance. O'Shaughnessy et al. [1] document the pattern across systems worldwide and show that solar curtailment rises sharply once penetration passes a threshold that is reached quickly where network expansion lags. In Chile's National Electricity System (Sistema Eléctrico Nacional, SEN), recorded curtailment rose from about 1.5 TWh in 2022 to more than 6 TWh per year by 2024, and the cumulative figure over 2022 to mid-2026 reaches roughly 19 TWh. For plant owners this is energy that is generated but never remunerated.

Anticipating curtailment is therefore commercially relevant, and it is harder than forecasting generation. Generation depends mainly on weather and installed capacity. Curtailment depends on the state of the whole system: net load, transmission congestion, the marginal cost of energy and the operation of other assets. It is zero-inflated and heavy-tailed: on many days a plant curtails nothing, and on the days it does the amounts vary widely. A single predicted number conveys neither the probability that curtailment occurs nor the range it may take, so what a plant owner needs is a calibrated statement of uncertainty, for instance an upper prediction limit that is exceeded with a known probability.

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The Chilean case adds a further difficulty. The distribution of curtailment changed during the study period: a change-point analysis (Section 3.4) places a ramp of roughly eighteen months between mid-2023 and late 2024, during which the positive magnitudes roughly tripled. The change has no single identifiable cause, since solar growth behind a saturated north–south corridor, the entry of battery storage, transmission reinforcements and changes in mining demand overlap in time, and it is superimposed on a strong annual cycle. A model calibrated on one part of that trajectory is miscalibrated on a later part.

Adaptive conformal methods, such as the online update of Gibbs and Candès [2], are the standard response when exchangeability fails, and they are often evaluated by comparing an adaptive scheme with a static conformal split on one base model and reporting the width saved at nominal coverage. This paper shows that such a comparison can be misleading, and quantifies why. It makes three contributions.

The first is a dataset. We release an open, plant-level dataset of curtailment in the SEN at daily and hourly resolution, spanning 53 continuous months and 300 plants across solar, wind and two hydropower technologies, built from the public reports of the system operator, corrected for eight classes of errata and reconciled exactly with the official annual totals.

The second is an evaluation of adaptive conformal calibration under a documented regime change. With a lognormal hurdle as base model, an adaptive scheme that transports calibration scores onto the recent regime and recalibrates online narrows the intervals of the most divergent window by about a third at near-nominal coverage. We repeat every experiment over nine base predictives from two model families and find that (i) the gain does not transfer across base models; (ii) to first order it is a mechanical conversion of the static split's coverage error into interval width, which a constant-threshold oracle with the same coverage reproduces without any adaptation, and which a controlled simulation reproduces with known ground truth; and (iii) holding the evaluation window fixed, the static split's coverage error, and with it the apparent gain, depends on the base model. The apparent gain therefore measures how far the calibration of the base predictive drifts between windows, not how much distribution shift the layer absorbs. Three checks follow, which are cheap and apply to any base model: report the static split's coverage in the evaluation window, the width of the oracle re-levelled split, and the same comparison on a second base model.

The third is an operational assessment. We weight coverage by installed capacity and by curtailed energy, evaluate high-curtailment days defined both by the outcome and by information available at the forecast origin, and price the limits with a simple cost model in which the one-sided interval score is the expected cost of a plant owner. On these criteria the transport-based scheme is not a method we recommend; a flexible base predictive with a static conformal wrapper is.

The evidence comes from one power system, two model families and a seven-day horizon, and we state the corresponding limits where they apply and in Section 7. Section 2 reviews related work, Section 3 describes the dataset and dates the change, Section 4 presents the base models, the calibration layers and the evaluation, Section 5 reports the results, Section 6 discusses them, Section 7 lists the limitations and Section 8 concludes.

2. Related work

This work draws on the applied literature on curtailment and the network constraints that cause it, on hybrid physical and data-driven modelling of power systems, and on the methodology of distribution-free uncertainty quantification and forecast evaluation. Each reference is discussed for what it establishes and for where our problem departs from it.

2.1. Curtailment: causes, measurement and forecasting

Curtailment is best understood as the visible output of a constrained network rather than as a property of the plants that suffer it. Henriot [3] formalises the economically efficient level of curtailment and shows that some curtailment is optimal, which frames forecasting as anticipating a rational system response. Joos and Staffell [4] quantify the short-term integration costs of that response in Britain and Germany and find curtailment and balancing costs of comparable order, which is what makes anticipation commercially relevant. Tang et al. [5] document the Chinese experience, where solar curtailment triggered a national policy response, and attribute it mainly to transmission inadequacy. Schermeyer et al. [6] take the network view and show, on German data, how congestion management determines which plants are curtailed and when. Nycander et al. [7] decompose Nordic curtailment into transmission

capacity, inertia and flexibility limits, the closest published analogue to the corridor-limited situation studied here. Yildiz et al. [8] use metered data from low-voltage networks to show how distributed photovoltaics and behind-the-meter storage are curtailed in practice and how poorly aggregate statistics capture it.

Forecasting curtailment, as opposed to explaining it, has received less attention. Bunodiere and Lee [9] predict curtailment in Kyushu with a rule-based method built on the operator's dispatch logic, which works where that logic is published and is hard to transfer where it is not. Our problem differs: we forecast plant-level curtailment directly, seven days ahead, without access to the operator's dispatch model, and we are concerned with the uncertainty of the forecast rather than with its point accuracy.

2.2. Storage, congestion and network capacity

The physical mechanisms matter here because they are the candidate explanations for the distributional change we document and because they bound what a statistical forecast can know. Denholm and Mai [10] establish how much storage, and of what duration, is needed to absorb a given level of curtailment. Wang et al. [11] embed the question in unit commitment and show the trade-off between operating cost and wind curtailment when batteries participate in dispatch. Rayit et al. [12] evaluate the storage owner's economics when curtailed wind is the input to grid services. Mohamad et al. [13] optimise where storage is placed so as to minimise solar curtailment under power-flow constraints, which explains how storage in one region can fail to relieve curtailment in another.

On the network side, Vargas et al. [14] study wind curtailment and storage as instruments of congestion management, including ramp-rate limits, on a corridor-constrained system similar in geometry to the SEN. Fang et al. [15] characterise locational marginal prices under jointly uncertain renewable generation and demand, the price signal behind the dispatch decisions that produce curtailment. A related line of work relaxes the constraint itself by rating network elements dynamically. Foss et al. [16] give the original dynamic ampacity algorithm. Teh and Cotton [17] show that dynamic thermal rating in a wind-integrated network changes reliability, not only capacity, because rating and wind resource are correlated. Lai and Teh [18] review the field and its practical limits. Glaum and Hofmann [19] quantify the capacity that dynamic line rating would unlock on the German grid, and Su et al. [20] forecast short-term transmission capacity under dynamic line rating. For cable-limited corridors the binding constraint is thermal: Maximov et al. [21] show how non-uniform soil temperature changes cable ampacity, and Ocłoń [22] relates soil thermal conductivity to material cost. We do not model these mechanisms; a forecaster with access to them would be solving a better-posed problem than ours.

2.3. Hybrid physical and data-driven modelling

A common response to a data-driven model that drifts is to anchor it in physics. Lei et al. [23] embed power-flow structure in a learned optimal power flow and show that the physical constraints improve extrapolation. Kruse et al. [24] apply the idea to grid frequency, where the dynamics are known and the disturbances are not. Kirchner-Bossi et al. [25] combine a physical wind-farm flow model with a data-driven correction and find that the hybrid beats either component. This route is closed to us because the physics that determines curtailment is the operator's security-constrained dispatch, and neither the network model nor the constraint set is public. Our base models are therefore purely statistical, and the calibration layer is asked to cope with the shift.

2.4. Probabilistic forecasting and conformal prediction

Probabilistic forecasting in power systems relies largely on quantile methods. Maciejowska et al. [26] introduce factor quantile regression averaging for electricity prices, and Uniejewski and Weron [27] regularise it and show that the gain is robust. Van der Meer et al. [28] study prediction intervals for solar power and net load and document how coverage degrades with aggregation level and penetration, the same conditional-coverage question we face across plants of different size. Mangalova and Shesterneva [29] show that a nearest-neighbour construction is competitive in the GEFCom2014 probabilistic wind track, a reminder that simple non-parametric predictive distributions are hard to beat. None of these methods carries a finite-sample coverage guarantee.

Conformal prediction supplies one. Angelopoulos and Bates [30] give the modern account: marginal coverage that holds in finite samples and without distributional assumptions under exchangeability, which is the assumption a regime change violates. Tibshirani et al. [31] extend the guarantee to covariate shift by reweighting calibration points with a density ratio, which presupposes an invariant conditional law $\mathbb{P}(Y \mid X)$. Barber et al. [32] drop exchangeability,

using fixed non-uniform weights and bounding the coverage gap in total variation; the bound covers regime change but cannot be computed from data. Gibbs and Candès [2] update the miscoverage level online from observed errors and attain long-run coverage under arbitrary shifts; their scheme is the adaptive component we use. Zaffran et al. [33] study the same scheme on time series, including electricity prices, show that its behaviour depends strongly on the step size and propose aggregating several step sizes; their finding motivates the per-model selection of the step in Section 4.7. Guan [34] localises the calibration set around the test point, an alternative route to conditional validity that we do not pursue.

Two constructions are central to our implementation. Chernozhukov et al. [35] score observations by their probability integral transform, which makes scores comparable across plants of very different size and handles the atom at zero; our score is theirs. Romano et al. [36] conformalize quantile regression directly, measuring nonconformity in the units of the response; we apply their method, in one-sided form, to every base model. Applications to power systems are growing. Hu et al. [37] conformalize a temporal convolutional quantile network for wind power intervals, and Jensen et al. [38] show that ensembling conformalized quantile regressors improves coverage and sharpness for time series. Neither addresses regime change or asks how much of the benefit attributed to a calibration layer depends on the base model it corrects.

Three further tools are used rather than extended. Peyré and Cuturi [39] provide the computational treatment of optimal transport from which we take the monotone one-dimensional map; Chernozhukov et al. [40] show that rearranging crossing quantile curves into monotone ones does not worsen them, which licenses the rearrangement step of our quantile base models; and Killick et al. [41] give the PELT algorithm used to date the distributional change.

2.5. Evaluating interval forecasts for decisions and under dependence

Gneiting and Raftery [42] give the theory of proper scoring rules from which our primary metric, the one-sided interval score, is derived. Gneiting and Ranjan [43] introduce threshold- and quantile-weighted versions of the continuous ranked probability score that emphasise a region of interest, such as the upper tail, while remaining proper; we use the threshold-weighted form to evaluate the base predictives in the tail. Lerch et al. [44] show that evaluating forecasts only on the cases in which an extreme outcome occurred, the forecaster's dilemma, rewards forecasters who overpredict extremes and can rank forecasts wrongly; this is why we report coverage on high-curtailment days defined both by the outcome and by information available when the forecast is issued. Our comparisons rest on a small number of base models and windows that share one panel. Cameron, Gelbach and Miller [45] show that a wild cluster bootstrap gives reliable inference with few clusters where cluster-robust standard errors do not; MacKinnon and Webb [46] extend the analysis to clusters of very different sizes; and Webb [47] proposes six-point weights for the case of very few clusters. We use these tools, together with a day-block bootstrap, to attach uncertainty to the cross-model relationships in Section 5.4.

3. Data

3.1. Source and scope

The dataset consolidates the curtailment records published monthly by Chile's National Electricity Coordinator (Coordinador Eléctrico Nacional, CEN) into two validated relational tables. The daily table contains 293,678 records and the hourly table 6,906,761 records, both covering the continuous period from 1 January 2022 to 31 May 2026, that is 1,612 days built from all 53 monthly reports, with no missing days. Records span 300 generation plants across four technologies: solar photovoltaic (75 plants), wind (56), run-of-river hydropower (140) and reservoir hydropower (29). Twelve hydropower plants are labelled inconsistently across the operator's monthly reports, alternating between the run-of-river and reservoir categories; the plant registry resolves each of them to a single canonical technology, so that a count of distinct plant-technology pairs in the raw daily table returns 312 rather than 300. Hydropower curtailment records are present from June 2024 onward, with hourly hydropower detail from July 2024, following the coverage of the underlying public reports.

Each plant is linked to a metadata registry matched to the CEN installation catalogue, with 18 fields including the official identifier, installed capacity, owner, region, province, municipality, coordinates, commissioning date, connection point and regulatory classification; the matching method is recorded for every plant. Days and hours without

curtailment are stored explicitly as zero-valued records, which represent 49.8% of daily and 83.6% of hourly rows. The explicit encoding of zeros preserves the zero-inflated structure of the quantity, which the models of Section 4 depend on. Figure 1 shows the geographic distribution of the plants.

3.2. Construction and validation

The 53 monthly reports were retrieved from the CEN public portal with a versioned download script, parsed and loaded into a relational database, and the original files were preserved unmodified with SHA-256 checksums. During construction we identified eight classes of errata in the original publications. Five were corrected or recovered deterministically in the curated tables and three are limitations of the source that we document without correction. Among the corrections, two omitted calendar days were recovered from the corresponding monthly hourly sheets and cross-checked to the third decimal against the source files' own total columns; one of them was detectable only by a structural continuity check, since the omitted day carried zero energy and was therefore invisible to any energy-based validation.

Two independent checks establish the fidelity of the result. Annual totals from the daily table match the CEN official summary exactly, to 0.000 GWh, in every year from 2022 to 2026, and aggregate 2022 wind and solar curtailment reproduces the 1.47 TWh independently reported by Fraunhofer Chile for that year. At monthly resolution the agreement is close but not exact: in a month-by-month reconciliation of hourly sums against daily totals over all 53 months, 50 months agree within 1% and 35 agree exactly, while the remaining three correspond to documented inconsistencies within the operator's own publications. The residual sub-percent differences arise from the operator assigning the same monthly energy differently across sister run-of-river units in its hourly sheets and in its annual closing, with a net effect of zero at system level. The dataset totals 19,030 GWh of recorded curtailment.

The dataset is publicly available on Zenodo under a CC BY 4.0 licence [48], together with the data dictionary, the errata log and the integrity checksums. The processing pipeline that regenerates every curated table is available in the companion code repository.

3.3. Exploratory characterisation

Three features of the data govern the modelling choices that follow.

First, curtailment is strongly zero-inflated and its occurrence is persistent, with a daily lag-one autocorrelation of roughly 0.85 at system level and a marked weekly cycle. Second, the positive magnitudes are heavy-tailed and approximately lognormal, with a ratio between the 99th and 50th percentiles of about twenty-five across technologies and about fifteen for solar alone (Figure 2).

Third, the change in the distribution of curtailment acts on scale rather than on occurrence or timing. For solar plants in the northern regions, the probability of a positive day averaged over each evaluation window stays between 0.750 and 0.815 from January 2024 to May 2026, and a change-point search on the monthly series of $\mathbb{P}(Y > 0)$ returns no break after September 2022; at monthly resolution the series ranges more widely, from 0.635 to 0.877. The intraday profile is also stable: its peak is at hour 16 in every half-year from 2022 to the first half of 2025 and at hour 17 afterwards, two adjacent hours whose profile masses differ by less than 0.4 percentage points, and the total variation distance between consecutive half-years stays between 0.032 and 0.079 after 2023. The positive magnitudes, in contrast, roughly triple: the median positive daily curtailment of solar plants in Antofagasta and Atacama rises from 92 MWh in the calibration window to 267 MWh in October to December 2024.

3.4. Dating the distributional change

The timing of the change in scale is best read after removing the annual cycle. Figure 3 shows the monthly mean of $\log Y$ over positive plant-days for solar plants in Antofagasta and Atacama, together with the same series net of a calendar-month median. Applied to the deseasonalised series with the PELT algorithm [41] and with binary segmentation, both under an L_2 cost and a BIC penalty $\hat{\sigma}^2 \log n$ with a minimum segment length of three months, the two algorithms return the same two breaks: May 2023, with a jump of +1.05 in mean log-magnitude, and January 2024, with a jump of +0.76. Neither returns a break in October 2024 or in 2025. Across 180 detection configurations (two strata, three deseasonalisation methods, five penalty multipliers and three minimum segment lengths) no configuration returns a break in 2025 or in October 2024. The change is therefore a ramp of roughly eighteen months, essentially

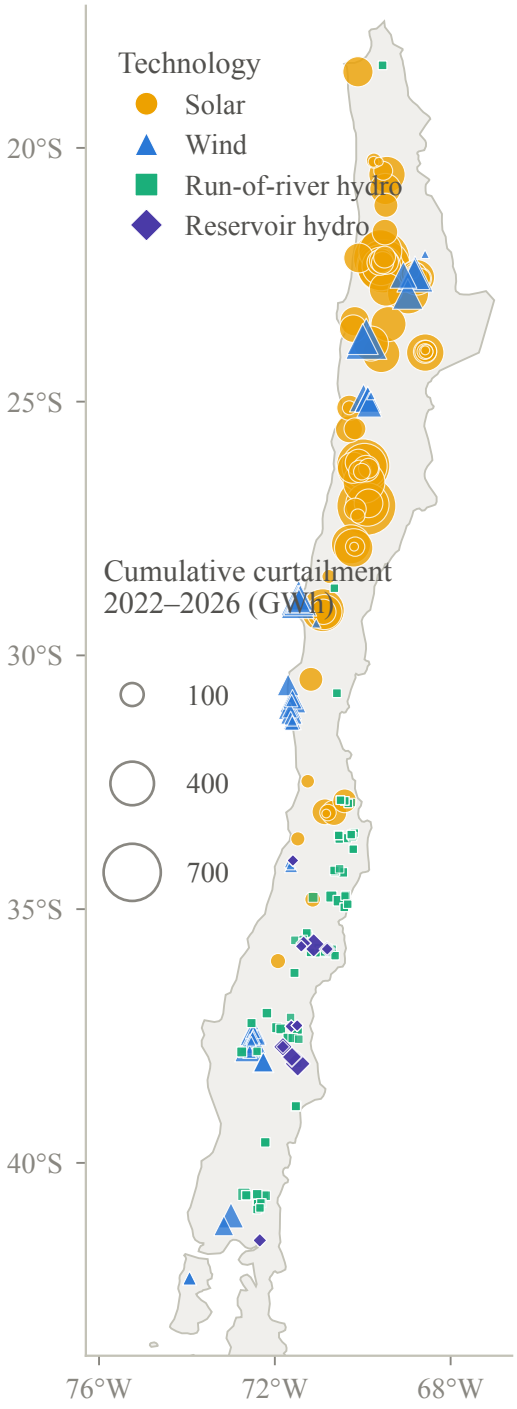


Figure 1: Geographic distribution of the 300 power plants in the dataset, coloured and shaped by technology, with marker area proportional to each plant's cumulative curtailed energy over January 2022 to May 2026.

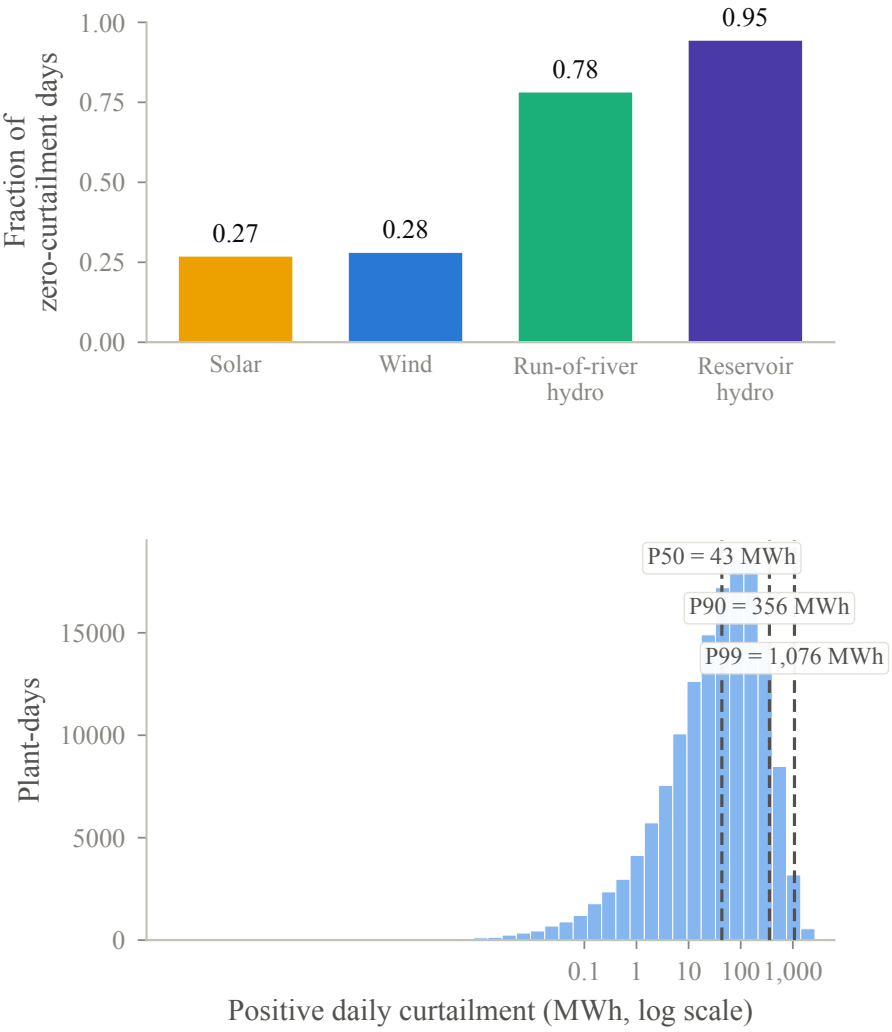


Figure 2: Distributional structure of plant-level curtailment. Top: share of zero-valued plant-days by technology. Bottom: distribution of positive magnitudes on a logarithmic scale, with the 50th, 90th and 99th percentiles marked.

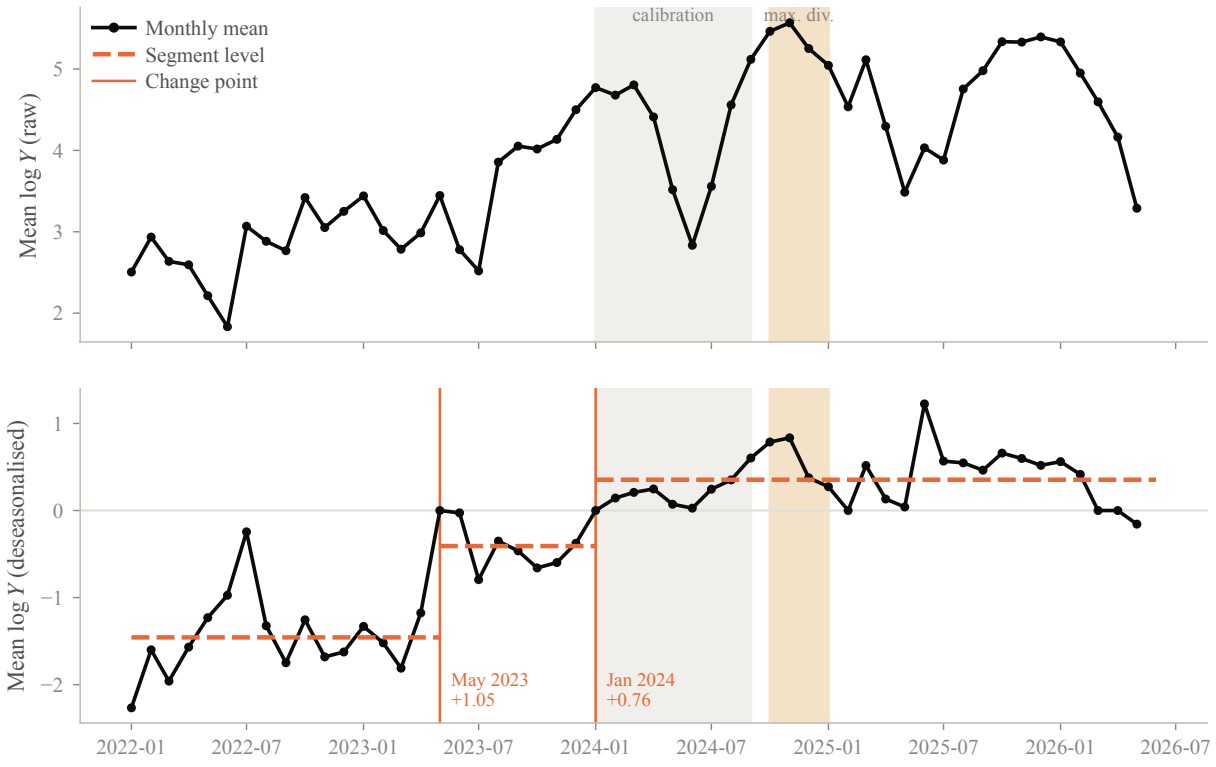


Figure 3: Monthly mean of $\log Y$ over positive plant-days, solar plants in Antofagasta and Atacama. Top: raw series, dominated by the austral annual cycle. Bottom: the same series net of a calendar-month median, with the two change points returned by PELT and binary segmentation (May 2023 and January 2024) and the calibration and transition windows of Section 4 shaded. The deseasonalised level peaks in October and November 2024 and then stops rising; the isolated spike of June 2025 is discussed in Section 6.

complete by the end of 2024: the Wasserstein-1 distance between the positive log-magnitudes of October to December 2023 and of October to December 2024 is 1.20, whereas between October to December 2024 and October to December 2025 it is 0.085, below the 0.115 threshold of a permutation null that resamples whole days.

Because the annual cycle is strong, contrasts between windows in different seasons register large distances even without a regime change. Table 1 decomposes the January–September to October–December shift in the monthly mean of $\log Y$ into a seasonal component, identical in every year, and a secular remainder. In 2024 the shift is the largest of the record, but 61% of it for northern solar, and 76% at system level, is the recurring seasonal component; the secular remainder of 2024, +0.46 for northern solar, is the largest of the four years but of the same order as the +0.29 of 2023. The raw contrast is used throughout the results, and should not be read as if it were entirely a regime effect.

4. Methods

4.1. Problem setup and base forecasters

For each plant i and target day T we forecast the curtailed energy $Y_{iT} \geq 0$ from information available at the forecast origin $t = T - 7$. The feature vector x_{iT} has twelve components: the most recent observation $y_{i,t}$, the lags $y_{i,t-1}$, $y_{i,t-7}$ and $y_{i,t-14}$, seven- and thirty-day trailing means ending at t, an occurrence indicator $\mathbb{1}\{y_{i,t} > 0\}$, the day of week and day of year of the target, the log of installed capacity, and region and technology as categorical variables. Every dynamic feature is a shift of at least seven days on a calendar-continuous index, verified by assertion when the

Table 1: Decomposition of the January–September to October–December shift in the monthly mean of log Y over positive plant-days, in log units, by year. The seasonal component is the difference between the calendar-month effects of the two windows and is by construction the same in every year; the secular component is the remainder. The 2024 row is the contrast that defines the transition window used in the evaluation; the final row of each block repeats the decomposition on the calibration window itself (January to August against October to December 2024), which is the pair the evaluation actually uses.

Year	Total shift	Seasonal	Secular	Seasonal share
Solar, Antofagasta and Atacama				
2022	+0.64	+0.72	−0.09	113%
2023	+1.01	+0.72	+0.29	72%
2024	**+1.18**	**+0.72**	**+0.46**	61%
2025	+0.90	+0.72	+0.17	81%
2024, calibration window	+1.29	+0.78	+0.50	61%
System (solar and wind)				
2022	+0.77	+0.66	+0.11	86%
2023	+0.89	+0.66	+0.23	74%
2024	**+0.86**	**+0.66**	**+0.20**	76%
2025	+0.71	+0.66	+0.05	93%
2024, calibration window	+0.94	+0.69	+0.24	74%

Note: the calendar-month effects are estimated as the median of each calendar month over the whole record. A robust STL decomposition returns somewhat different values on a 53-month series carrying a strong trend; it does not change the order of magnitude of the split, and the secular components remain the smaller part of the 2024 contrast.

features are built, which rules out leakage. The modelling panel contains the 131 solar and wind plants. All base models are fitted on targets up to 31 December 2023 and never see the evaluation period.

We use nine base predictives from two model families.

Lognormal hurdle. The predictive factorises into occurrence and magnitude,

$$Y \mid x \;\sim\; (1 - p(x))\,\delta_0 \;+\; p(x)\,\mathrm{LogNormal}(\mu(x), \sigma^2), \tag{1}$$

with $p(x)$ from a gradient-boosting classifier and $\mu(x)$ from a gradient-boosting regression on $\log y$ over positive days. The dispersion σ is estimated out of fold by five-fold cross-fitting on the training period, which avoids the downward bias of the in-sample residual variance of a boosted model. Four variants are used. The *reference hurdle* has two models of 400 trees each, with hyperparameters fixed a priori (learning rate 0.05, depth 6, minimum child weight 5, row and column subsampling 0.9), and $\hat{\sigma} = 1.70$. The *early-stopped hurdle* holds out the last six months of the training period (July to December 2023) as a temporal validation set, fits each stage to a ceiling of 4,000 trees with a patience of 50 rounds, and refits both stages on the full training period at the selected budget: 59 trees for occurrence and 46 for magnitude, 105 in total, with $\hat{\sigma} = 1.83$. The *conditional-dispersion hurdle* replaces σ by a fitted $\sigma(x)$, estimated out of fold from the squared stage-one residuals with a global scale correction and a floor at the fifth percentile of the training distribution. The *enlarged hurdle* has two models of 2,000 trees.

Multi-quantile gradient boosting. A predictive defined by a dense grid of 54 estimated quantiles, at levels 0.02 to 0.98 in steps of 0.02 plus 0.005, 0.01, 0.99, 0.995 and 0.999, fitted with the pinball loss under the same features, rows, training cut-off, per-model hyperparameters and seed as the hurdle. Crossing quantile curves are rearranged into monotone ones [40] by sorting each row and clipping at zero. The predictive CDF is the interpolated inverse of the quantile function, so it is not committed to any tail shape, and the mass at zero is read off the grid as $F_x(0) = \max\{\tau_k : \hat{q}_{\tau_k}(x) = 0\}$. Above the highest level, $\tau = 0.999$, the predictive expresses nothing, and a requested quantile above it returns $+\infty$. Five variants differ only in the number of trees per quantile level: 15, 50, 100, 200 and 400. The variant with 400 trees per level is the *reference quantile model*.

Three base models appear in every main comparison: the reference hurdle, the early-stopped hurdle and the reference quantile model. The other six complete the cross-model analysis of Section 5.4.

4.2. Conformal calibration with a PIT score

We score each observation by its probability integral transform under the predictive law,

$$s \;=\; F_x(y), \tag{2}$$

the distributional-conformal score of Chernozhukov et al. [35]. Because F_x has an atom at zero, the score is randomised there in the standard way for mixed distributions [49]: for $y = 0$ we set $s = F_x(0^-) + U(F_x(0) - F_x(0^-))$ with $U \sim \text{Uniform}(0, 1)$, which for the hurdle reduces to $s = (1 - p(x))\,U$. Under the true predictive law s is Uniform(0, 1), on a common scale across plants of any size. The calibration layer asks only two operations of the base model, the CDF and its generalised inverse, so every base model enters through the same interface and every comparison across base models changes nothing else.

Given a calibration set of size n, the conformal quantile is the order statistic $\hat{q} = s_{(k)}$ with $k = \lceil (n + 1)(1 - \alpha) \rceil$, which yields $\mathbb{P}(Y \le U(x)) \ge 1 - \alpha$ under exchangeability; when $k > n$ the quantile is $+\infty$. For the hurdle the upper prediction limit $U(x) = F_x^{-1}(\hat{q})$ is zero when $\hat{q} \le 1 - p(x)$, $+\infty$ when $\hat{q} \ge 1$, and otherwise

$$U(x) \;=\; \exp\Big(\mu(x) + \sigma\,\Phi^{-1}\big(\tfrac{\hat{q} - (1 - p(x))}{p(x)}\big)\Big), \tag{3}$$

as shown in Appendix A. For the quantile model the inversion is a linear interpolation on the quantile grid.

For the hurdle, a change of scale c of the positive magnitudes is a translation by $\log c/\sigma$ on the latent Gaussian scale of the score, that is, a monotone displacement of the law of s on $[0, 1]$. This structure motivated the transport step described below. It belongs to the hurdle, whose positive magnitudes share one dispersion; for the other base models the transport is the same algorithm without that motivation.

4.3. Three guarantees, and which one applies where

Three statements are easy to conflate and are kept separate in this paper. *Finite-sample marginal validity*: the static split satisfies $\mathbb{P}(Y \le U(x)) \ge 1 - \alpha$ at the stated n if the calibration scores and the test score are exchangeable; this is an average over draws of the calibration set and the test point, not a statement about any plant, day or window. *Long-run adaptive control*: the online update of Gibbs and Candès [2] guarantees $\frac{1}{T}\sum_{t \le T} \text{err}_t \to \alpha$ for any sequence, without distributional assumptions, but says nothing about coverage at a finite time. *Empirical coverage in this panel*: what we measure, which neither statement describes, because exchangeability fails across the regime boundary and also within it. All plants share the daily state of the system, so scores arrive in correlated daily blocks: on the calibration set the intraclass correlation of the PIT score by date is 0.238, and with 108.8 plants per day the design effect is 26.7, so the effective calibration sample is about 1,000 rows rather than 26,552. Every coverage figure therefore carries a standard error clustered by date, and every difference between methods a bootstrap interval that resamples whole days.

4.4. Calibration layers

All layers produce a one-sided upper limit $[0, U]$ at nominal level $1 - \alpha = 0.90$ and use the same calibration window, January to August 2024.

Static split. The conformal quantile of the calibration scores, fixed for the whole test period.

Sliding window. The conformal quantile of the scores of the last 60 days, refreshed every 7 days. Because the forecast horizon is 7 days, the outcome of an origin is known only 7 days later, so every online window ends 7 days before its target date; this horizon embargo applies to all online layers.

Adaptive conformal inference (ACI). The miscoverage level is updated online [2], $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, with err_t the fraction of plants of day t whose outcome exceeds its limit, fed back after the 7-day embargo. Left unbounded, the update can drive α_t to zero, at which point the conformal quantile does not exist and the limit is $+\infty$; we report both the unbounded update and an update bounded below at $\alpha_{\min} = 0.005$.

Transport + ACI. The calibration scores are mapped onto the distribution of the scores in a recent window by the monotone one-dimensional transport map $T = \hat{F}_{\text{new}}^{-1} \circ \hat{F}_{\text{old}}$ [39], the conformal quantile is taken from the transported pool at the current α_t, and α_t is updated as in ACI. In the upper tail, where the recent window is noisy, the map is shrunk towards the identity in proportion to a bootstrap estimate of its variance. Because the map depends on the data,

the transported scores are not exchangeable with the test score; coverage is controlled by the online update and the map contributes only sharpness (Appendix B).

One-sided conformalized quantile regression (CQR). Following Romano et al. [36], the nonconformity score is $E = y - q_{1-\alpha}(x)$, measured in MWh, where $q_{1-\alpha}(x) = F_x^{-1}(1-\alpha)$ is the base model's own quantile, and the limit is $U(x) = \max\{q_{1-\alpha}(x) + \hat{Q}_{1-\alpha}(E), 0\}$ with $\hat{Q}$ the conformal quantile of E on the calibration window. We apply it to every base model, so that the comparison between calibration layers is made on a common base model.

Weighted conformal prediction [31] is not used: the shift changes the conditional law $\mathbb{P}(Y \mid x)$, which no reweighting of old scores can reflect, and features that trend in time separate calibration from test almost perfectly, so the density-ratio estimate degenerates.

4.5. The transport-based scheme in full

Algorithm 1 gives the procedure, Figure 4 its data flow, and Appendix C every constant and seed. Three implementation choices are stated here because a reader would otherwise have to guess them.

Quantile grid and interpolation. The map is estimated on an equispaced grid of $g = \min(100, \max(20, m/2))$ probability levels, where m is the number of scores in the recent window, and applied by two successive linear interpolations, from a score to its level on the old empirical quantile function and from that level to the regularised new quantile.

Shrinkage. The recent window is bootstrapped $B = 150$ times; the map is the mean of the replicate quantile functions, and their standard deviation $\mathrm{sd}_{\mathrm{boot}}(u)$ at each level u measures its uncertainty. The regularised quantile is

$$\hat{q}_{\mathrm{reg}}(u) = \big(1 - \lambda(u)\big)\, \hat{q}_{\mathrm{old}}(u) + \lambda(u)\, \hat{q}^{\mathrm{bag}}_{\mathrm{new}}(u), \qquad \lambda(u) = \frac{1}{1 + \big(\mathrm{sd}_{\mathrm{boot}}(u)/\tau\big)^2}, \tag{4}$$

with $\tau = 0.05$ in PIT units, fixed a priori. Where the bootstrap is stable the map transports fully; where it is noisy it reverts to the old quantile. The result is monotonised by a running maximum and clipped to $[0, 1]$.

Order of execution. On date t: (i) if a refresh is due, the map is recomputed from the window ending at $t - 7$ and the whole calibration pool is transported; (ii) the conformal quantile is taken at the current α_t; (iii) the limit is issued; (iv) the error of day t enters the embargo queue and updates α seven steps later. The map never uses the adaptive level, and the update never uses the map.

4.6. What is taken from the literature and what is not

The PIT score is that of Chernozhukov et al. [35]; the randomisation of the atom is the device of Smith [49]; the online update is that of Gibbs and Candès [2]; the monotone one-dimensional map is the increasing rearrangement of optimal transport [39], which Chernozhukov et al. [40] apply to quantile curves; bagging and shrinking towards a stable reference in proportion to the bootstrap variance is ordinary shrinkage practice. What is specific to the scheme is the combination, a transport of the score distribution used as a sharpness-only step inside an online update that carries the coverage control, and the variance-proportional tail shrinkage of (4). Section 5.7 evaluates each component separately.

4.7. Hyperparameter selection

The step γ and the length of the recent window are selected separately for each base model by rolling-origin validation inside the calibration window. Base-model predictions start on 1 January 2024, so January to April 2024 serves as an inner calibration pool and May to August 2024 as an inner validation set, evaluated with the same seven-day refresh and embargo as the test protocol. The criterion is the mean one-sided interval score of Section 4.8, which is $+\infty$ if a configuration produces any infinite limit, so such configurations are excluded by the criterion itself. The grids are $\gamma \in \{0.005, 0.01, 0.02, 0.05, 0.10, 0.20\}$ and windows of $\{30, 45, 60, 90, 120\}$ days. The selected values are reported in Appendix C. The test period is used once, for the final evaluation; the full grid evaluated on the test period appears in Appendix E for information only. Base-model hyperparameters were fixed a priori, except for the number of trees of the early-stopped hurdle, which is selected on the training period as described above.

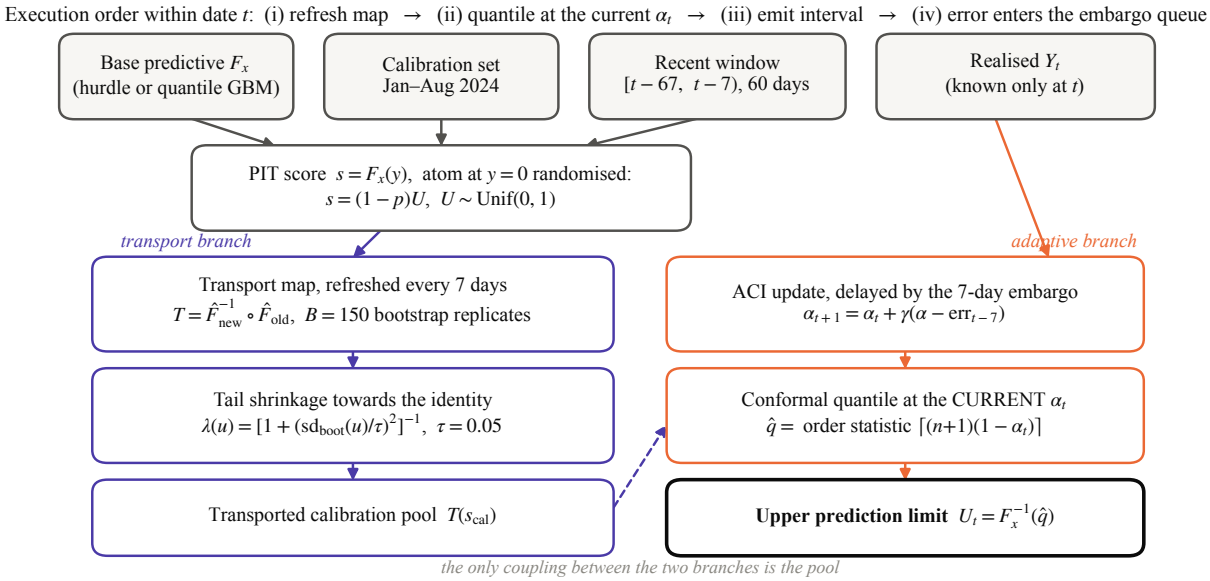


Figure 4: Data flow of the transport-based scheme, from the base predictive to the upper prediction limit. The transport branch (left) estimates and regularises the map; the adaptive branch (right) maintains the effective miscoverage level. The two are coupled only through the transported calibration pool; the order of operations within a date is stated at the top.

4.8. Evaluation

Windows. The calibration set is January to August 2024 (244 days, 26,552 plant-days). The test period, September 2024 to May 2026 (638 days, 77,868 plant-days), is split into five windows: September 2024; October to December 2024, the *transition window*; and the semesters 2025-S1, 2025-S2 and 2026-S1. The transition window is defined by a measurable criterion, maximum divergence from the calibration distribution: the Wasserstein-1 distance between its positive log-magnitudes and those of the calibration set is 1.25, against 0.28 to 0.95 for the other windows. That distance is almost entirely a displacement of location, and the seasonal decomposition of Table 1 attributes 61 per cent of it to the annual cycle for northern solar and 74 per cent at system level. The window is the hardest case the calibration layer faces in this record rather than the signature of a single event; Appendix G shows that the conclusions do not depend on where its boundaries are drawn.

Primary metric. The forecasts are one-sided limits $[0, U]$ at level $1-\alpha$, scored by the one-sided interval score

$$\mathrm{IS}_\alpha(U, y) \;=\; U \;+\; \frac{1}{\alpha}\,\max(y - U,\, 0), \tag{5}$$

which is $1/\alpha$ times the quantile (pinball) loss of level $1-\alpha$ plus a term that does not depend on U, and is therefore proper for that quantile [42]. It is in MWh, penalises width and non-coverage on the same scale, and is $+\infty$ whenever U is. Mean width over finite limits flatters a method that escapes into infinite limits, so we report three quantities: the interval score over finite limits, the unrestricted mean (infinite if any limit is infinite), and the fraction of infinite limits.

Operational reading. Consider a plant owner who must commit, seven days ahead, an allowance U for curtailment, paying c_h per MWh of allowance (for instance, energy that is not sold forward) and c_p per MWh curtailed beyond the allowance (energy that must be bought back or settled). The expected cost $c_h\,U + c_p\,\mathbb{E}\max(Y - U, 0)$ is minimised by the quantile of level $1 - c_h/c_p$, and in units of c_h it equals the one-sided interval score at $\alpha = c_h/c_p$. The interval score at $\alpha = 0.10$ is therefore the cost of an owner with $k = c_p/c_h = 10$. We also evaluate limits calibrated at $\alpha = 0.05$ and $\alpha = 0.01$, which correspond to $k = 20$ and $k = 100$; these two levels are also our tail-focused scores.

Algorithm 1 Transport + ACI with tail shrinkage. Constants and seeds in Appendix C.

Require: base predictive F_x; calibration scores s_{cal} of size n; nominal level α; step γ; window W; refresh period R; embargo E; bootstrap size B; shrinkage scale τ; bounds $\alpha_{\min}, \alpha_{\max}$

$\alpha_1 \leftarrow \alpha$; $\mathcal{P} \leftarrow \text{sort}(s_{\text{cal}})$; queue $Q \leftarrow \emptyset$; $t_{\text{next}} \leftarrow t_0$
for each test date t **do**
  if $t \geq t_{\text{next}}$ **then**
    $s_{\text{new}} \leftarrow$ scores with date in $[t - W - E,\ t - E)$
    if $|s_{\text{new}}| \geq 100$ **then**
      draw B bootstrap resamples of s_{new}; set $\hat{q}^{\text{bag}}_{\text{new}}$ to their mean quantile function and sd_{boot} to its replicate standard deviation
      form $\hat{q}_{\text{reg}}$ by (4); monotonise by running maximum; clip to $[0, 1]$
      $T \leftarrow$ interpolation from $\hat{q}_{\text{old}}$ to $\hat{q}_{\text{reg}}$; $\mathcal{P} \leftarrow \text{sort}(T(s_{\text{cal}}))$
    end if
    $t_{\text{next}} \leftarrow t + R$
  end if
  $k \leftarrow \lceil (n+1)(1-\alpha_t) \rceil$; $\hat{q}_t \leftarrow \mathcal{P}_{(k)}$ if $k \leq n$, else $+\infty$
  $U_t \leftarrow F_x^{-1}(\hat{q}_t)$ for every plant of date t ▷ limit issued
  push $\text{err}_t = \frac{1}{|I_t|} \sum_{i \in I_t} \mathbb{1}\{Y_{it} > U_{it}\}$ onto Q
  if $|Q| > E$ **then**
    pop err_{t-E} from Q; $\alpha_{t+1} \leftarrow \text{clip}(\alpha_t + \gamma(\alpha - \text{err}_{t-E}),\ \alpha_{\min},\ \alpha_{\max})$
  end if
end for

Weighted coverage. Plant-day coverage weights a 5 MW plant on a quiet day like a 300 MW plant on the most congested day. We therefore also report coverage weighted by installed capacity, coverage weighted by curtailed energy, $\sum_{i,t} y_{it} \mathbb{1}\{y_{it} \leq U_{it}\} / \sum_{i,t} y_{it}$, the share of curtailed energy left uncovered, $\sum_{i,t} \max(y_{it} - U_{it}, 0) / \sum_{i,t} y_{it}$, and the mean exceedance given that the limit is exceeded.

High-curtailment days. A day is high if system curtailment, summed over the plants of the panel, is high. We use three definitions: (H1) the day's curtailment is above the 90th percentile of daily system curtailment in the calibration window, which marks 254 of the 638 test days because the series rose; (H2) the day is in the top decile of the test period; (H3) the trailing seven-day mean of system curtailment known at the forecast origin, that is, ending seven days before the target, is above the same threshold. H1 and H2 condition on the outcome; H3 conditions only on information available when the forecast is issued. Evaluating forecasts only where the outcome turned out extreme favours forecasts that overpredict [44], so the outcome-based definitions describe exposure while H3 is the legitimate test of conditional calibration.

Base predictive. The continuous ranked probability score (CRPS), its threshold-weighted version [43] with weight $\mathbb{1}\{z \geq r_i\}$, where r_i is the 90th percentile of plant i's daily curtailment in the calibration window, and the Brier score of $\{Y > 0\}$ describe the base model and cannot arbitrate between calibration layers on the same base model.

Uncertainty. Coverage carries a standard error clustered by date. Differences between methods carry a 95% percentile interval from 2,000 bootstrap replicates that resample whole days.

4.9. Decomposing an adaptive gain

For a base model m and a window w, let C^{st}_{mw} and W^{st}_{mw} be the coverage and mean width of the static split, and C^{ad}_{mw} and W^{ad}_{mw} those of an adaptive layer. The quantities analysed across cells are the static over-coverage $\text{OC}_{mw} = C^{\text{st}}_{mw} - 90$, in percentage points, and the width change $\Delta_{mw} = 100\,(W^{\text{ad}}_{mw} / W^{\text{st}}_{mw} - 1)$, in per cent.

An adaptive layer that targets nominal coverage must narrow its limits when the static split over-covers and widen them when it under-covers, whatever the source of the coverage error. To separate this mechanical effect from adaptation proper, we compute for each cell an oracle: the constant PIT threshold q^*_{mw}, chosen after the fact, at which the static limit $F_x^{-1}(q^*_{mw})$ attains in window w exactly the coverage C^{ad}_{mw} of the adaptive layer. The oracle does not adapt in time; it only re-levels the static split. Its width change $\Delta^{\text{mec}}_{mw} = 100\,(W^*_{mw} / W^{\text{st}}_{mw} - 1)$ is the *mechanical* component, and the residual $\Delta_{mw} - \Delta^{\text{mec}}_{mw}$ measures what adaptation in time adds, or costs, relative to the best constant re-levelling

at the same coverage. A negative residual would indicate that the layer tracks variation within the window; a positive one, that it pays for fluctuations of its level. The oracle uses the outcomes of the window and is a benchmark, not a method.

The cells share days, base models and windows, so they are not independent. We attach uncertainty in three complementary ways. First, a day-block bootstrap draws whole days with replacement once per replicate and recomputes every cell, including the oracle, on that draw, which preserves the dependence between cells that share days. Second, we fit the cross-cell regression with fixed effects for windows, for base models and for both, and with crossed random effects. Third, we test the slope with a restricted wild cluster bootstrap clustered by base model and, separately, by window, following Cameron et al. [45] and MacKinnon and Webb [46], with the six-point weights of Webb [47], which remain usable with five clusters. None of these makes the two model families a sample from a population of model families; statements about other families rest on the mechanism, not on these intervals. Finally, the early-stopped hurdle is held out: the relationship is estimated on the other eight base models and then used to predict its five cells.

4.10. Controlled simulation

To separate the sources of a static split's coverage error with known ground truth, we simulate a zero-inflated panel of 100 plants observed for 240 calibration days and 638 test days split into five windows of the same lengths as the real ones. Occurrence follows a logistic model with plant effects and a common daily shock, and log-magnitudes follow a normal model with plant effects, a level path, a common daily shock and idiosyncratic noise with a time-varying standard deviation; both common shocks are first-order autoregressive, which induces the dependence across plants. Four scenarios specify the test period: no shift; a tripling of magnitudes, introduced as a ramp over the transition window; the same tripling together with a reduction of the idiosyncratic dispersion by 40 per cent; and the reduction of dispersion alone. Eight base predictives, all lognormal hurdles, span the relevant kinds of misspecification: a correctly specified oracle, constant over-dispersion by 30 per cent, partial tracking of the dispersion, a constant dispersion fixed at its calibration value with and without a constant scale error, and a level that lags the truth by 30 days with and without constant dispersion. Each replicate produces forty model-window cells, like the real analysis, which are passed through the same calibration code. Appendix F gives the full specification.

5. Results

5.1. Point forecasting, within the scope tested

Table 2 reports the mean absolute error of the point forecasts on the out-of-sample period. The seasonal-naive predictor, which averages the same weekday over the previous four weeks, is the most accurate, and the medians of the gradient-boosting models do not beat it; the median of the hurdle mixture, which is fitted for the whole predictive law, is the least accurate, and the early-stopped hurdle, whose predictive is better by CRPS (Section 5.6), is worse still by this criterion. The ordering follows from the strong weekly persistence of curtailment (Section 3.3). The conclusion holds for these model classes, this feature set, this seven-day horizon and this criterion; a model with access to the network model, weather forecasts, unit-commitment schedules or storage state of charge would be solving a better-posed problem, and a criterion that weights large days might order the same models differently. Within the information available to a plant owner from public data, effort is better spent on quantifying uncertainty than on refining the predicted number.

5.2. With the reference hurdle, adaptation appears to recover a third of the width

With the reference hurdle as base model, the shift does not break the coverage of the static split; it inflates it. Over the test period the static split covers between 89% and 95% in every window (Table G.16 in Appendix G), and the cost of the shift is paid in width, concentrated in the transition window, where the static split reaches 95.3% coverage with a mean width of 1203 MWh. Figure 5 traces the drift. Transport+ACI with $\gamma = 0.05$ brings the transition-window limits down to 806 MWh at 90.9% coverage, a reduction of 33% at a coverage within one point of nominal. Read alone, this is the result that motivated the scheme. Over the whole test, however, the same configuration returns infinite limits on 5.9% of plant-days, and its interval score over finite limits, 813 MWh, is not distinguishable from the 826 MWh of the static split. The next subsections ask what the one-third reduction measures.

Table 2: Mean absolute error (MWh) of the base forecasters on the out-of-sample period, on the common row set. Lower is better. The comparison is between these model classes on this feature set at this horizon under this criterion; see the scope statement in the text.

Model	2024	2025	2026	Total
Seasonal-naive (weekly)	84.3	86.3	84.2	85.1
Multi-quantile GBM (median)	87.7	89.0	84.1	87.6
Quantile GBM, 3 levels (median)	88.5	89.9	84.3	88.3
Persistence	88.4	96.3	90.7	92.2
Point XGBoost	92.7	94.6	89.7	93.0
Hurdle (mixture median)	104.4	104.8	101.6	104.1
Hurdle, early stopping (mixture median)	110.1	108.8	104.6	108.6

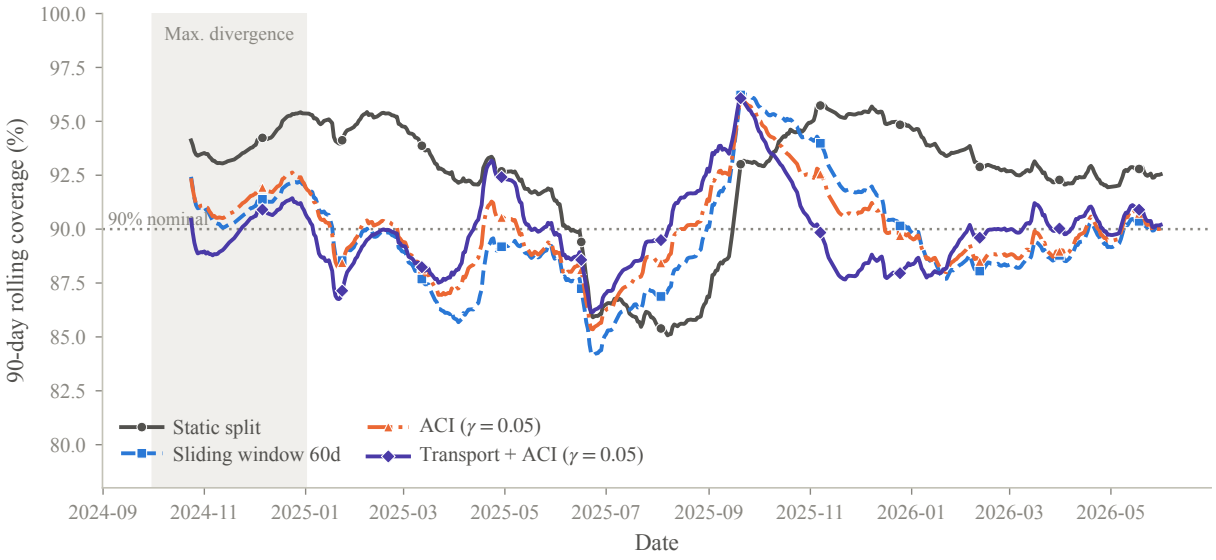


Figure 5: Rolling empirical coverage (90-day window) of the calibration layers with the reference hurdle as base model, with the 7-day horizon embargo. The dashed line marks the 90% nominal level and the shaded band the transition window, October to December 2024.

5.3. The same layer over different base models

Table 3 runs every calibration layer over the three main base models, and Figure 6(a) adds the conditional-dispersion hurdle. The calibration layer is identical across base models: same score, windows, embargo, seeds and constants.

The one-third reduction does not transfer. In the transition window the width change of Transport+ACI is −33.1% with the reference hurdle, −22.5% with the early-stopped hurdle, −4.8% with the conditional-dispersion hurdle and +3.6% with the quantile model, whose static split is already at 90.6% coverage. The static split's over-coverage in that window follows the same order: +5.3, +4.7, +1.7 and +0.6 percentage points.

The base model matters more than the layer. Over the whole test, the early-stopped hurdle with a plain static split has an interval score of 534 MWh, 35% below the static split on the reference hurdle and below every calibration layer on it, the best of which, one-sided CQR, reaches 714 MWh. Its static limits are 40% narrower than those of the reference hurdle at the same coverage (92.0% against 92.5%). On the early-stopped hurdle, adaptation adds little: the best online layer improves the interval score by 3%, Transport+ACI with $\gamma = 0.05$ worsens it by 3%, and the selected Transport+ACI configuration is within 1% of the static split. The quantile model is better again: its static split has an interval score of 433 MWh at 90.2% coverage.

Table 3: Calibration layers over the three main base models, nominal level 90%. Coverage in per cent with the date-clustered standard error, mean width over finite limits (MWh), one-sided interval score over finite limits (MWh) and fraction of infinite limits (%). In the whole-test interval score, an asterisk marks a method with infinite limits, whose unrestricted score is infinite. CRPS and twCRPS are those of the base predictive (MWh); the twCRPS weights the region above each plant's calibration-window 90th percentile. ΔIS: difference in whole-test interval score from the static split on the same base model, with a 95% day-block bootstrap interval, on plant-days where both limits are finite. No limit is infinite in the transition window. The selected configurations come from rolling-origin validation inside the calibration window (Appendix C).

	Transition window (Oct–Dec 2024)			Whole test (Sep 2024–May 2026)				
Method	Coverage	Width	IS	Coverage	Width	IS	% inf.	ΔIS vs. static [95% CI]
Reference hurdle (2×400 trees): CRPS 133.5 / 89.5, twCRPS 78.9 / 40.7								
Static split (PIT)	95.3 (0.45)	1203	1395	92.5 (0.37)	678	826	0.0	–
Sliding window, 60 d	92.0 (0.73)	834	1112	89.9 (0.44)	531	734	0.0	−93.4 [−109.8, −76.4]
ACI ($\gamma = 0.05$)	92.1 (0.67)	822	1096	90.1 (0.43)	544	753*	4.2	−95.9 [−117.7, −69.6]
Transport+ACI ($\gamma = 0.05$)	90.9 (0.81)	806	1107	90.1 (0.43)	596	813*	5.9	−32.3 [−69.8, +9.3]
ACI (selected γ)	94.8 (0.49)	1086	1294	90.9 (0.41)	590	763	0.0	−63.0 [−69.0, −57.0]
Transport+ACI (selected)	93.3 (0.54)	924	1162	90.0 (0.45)	564	761	0.0	−65.4 [−82.8, −47.9]
One-sided CQR	94.0 (0.55)	937	1175	91.2 (0.38)	532	714	0.0	−112.4 [−120.8, −103.7]
Early-stopped hurdle (105 trees): CRPS 124.0 / 84.4, twCRPS 70.3 / 37.2								
Static split (PIT)	94.6 (0.51)	689	823	92.0 (0.35)	409	534	0.0	–
Sliding window, 60 d	92.6 (0.69)	575	758	89.8 (0.39)	350	517	0.0	−16.6 [−22.4, −10.7]
ACI ($\gamma = 0.05$)	92.3 (0.61)	553	736	90.1 (0.37)	354	524*	3.3	−19.8 [−27.9, −11.0]
Transport+ACI ($\gamma = 0.05$)	91.1 (0.82)	534	748	90.0 (0.38)	377	552*	3.3	+9.4 [−7.8, +27.7]
ACI (selected γ)	94.3 (0.53)	655	797	90.8 (0.37)	377	520	0.0	−13.8 [−16.0, −11.6]
Transport+ACI (selected)	93.8 (0.52)	613	761	89.9 (0.40)	367	527	0.0	−6.6 [−12.9, −0.2]
One-sided CQR	96.2 (0.39)	840	932	94.2 (0.30)	501	590	0.0	+57.2 [+52.9, +61.6]
Reference quantile GBM (54×400): CRPS 93.0 / 68.0, twCRPS 51.6 / 28.4								
Static split (PIT)	90.6 (0.85)	410	588	90.2 (0.38)	288	433	0.0	–
Sliding window, 60 d	90.1 (0.94)	409	598	89.9 (0.41)	288	438	0.0	+5.7 [+4.0, +7.5]
ACI ($\gamma = 0.05$)	90.6 (0.87)	412	588	90.2 (0.41)	294	447*	2.8	+8.5 [+5.5, +11.4]
Transport+ACI ($\gamma = 0.05$)	90.5 (0.99)	425	602	90.2 (0.44)	305	459*	4.4	+19.1 [+14.1, +24.2]
ACI (selected γ)	90.6 (0.86)	411	588	90.3 (0.39)	289	434	0.0	+0.8 [+0.4, +1.1]
Transport+ACI (selected)	89.9 (0.95)	411	598	90.1 (0.41)	294	444	0.0	+11.0 [+8.3, +13.6]
One-sided CQR	89.3 (0.90)	389	602	89.8 (0.38)	276	438	0.0	+5.7 [+4.2, +7.2]

The ranking of the layers also depends on the base model. One-sided CQR is the best layer on the reference hurdle (714 MWh) and close to the best on the quantile model (438 MWh at 89.8% coverage), but on the early-stopped hurdle it over-covers (94.2%) and is worse than the static split (590 MWh). The reason is the scale of the nonconformity score. CQR corrects the base model's 0.9 quantile by a single number of MWh, the 90th percentile of $y - q_{0.9}(x)$ pooled over plants whose sizes differ by two orders of magnitude; on these data that percentile is dominated by small plants and zero days and lies close to zero, −1.1 MWh for the early-stopped hurdle. The early-stopped hurdle's own 0.9 quantile already covers 92.8% of the calibration window, and CQR leaves that over-dispersion in place, whereas the PIT score, which is free of scale, removes it by lowering the conformal level to 0.880.

5.4. How much of the gain is mechanical

Across the 40 cells of eight base models by five windows, the width change of Transport+ACI is linearly related to the static over-coverage, with $r = -0.87$ and a slope of −4.89% of width per percentage point (Figure 6b). The relationship predicts the five cells of the held-out early-stopped hurdle with a correlation of 0.96 between observed and predicted changes and a root mean squared error of 4.9 points. A relationship of this kind is expected from any layer that targets nominal coverage, so the question is how much of it requires adaptation at all.

The constant-threshold oracle answers it (Table 4 and Figure 7). Over the 45 cells, the oracle's width change has a slope of −4.80 against −4.98 for the observed change of Transport+ACI with $\gamma = 0.05$, and it explains 84% of the variance of the observed change. The residual, what adaptation in time adds beyond re-levelling, averages +6.9% with a 95% interval of [+5.3, +8.6] and is unrelated to the over-coverage ($r = -0.07$). The other configurations tell the same story. With the update bounded at $\alpha_{\min} = 0.005$ the mechanical slope is unchanged (−4.98), but the residual is larger and more variable (+11.2%), because when α_t reaches its bound the limits become very wide, a cost the oracle does not pay; this is why the oracle explains less of the variance in that configuration ($R^2 = 0.30$). Pure ACI

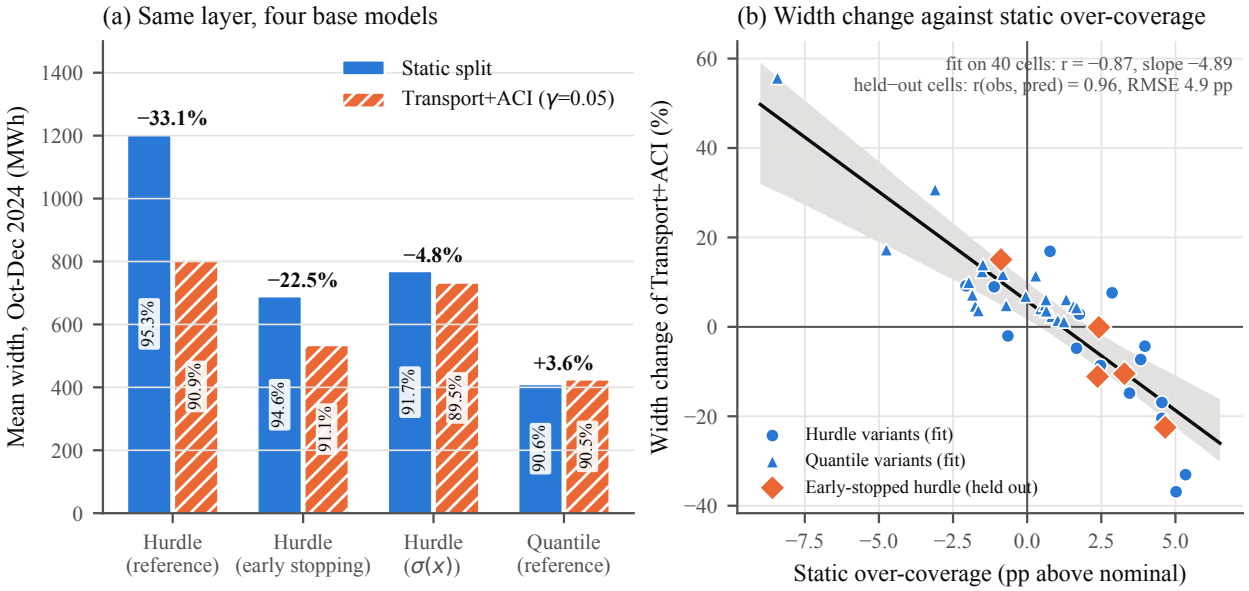


Figure 6: (a) Mean width in the transition window of the static split and of Transport+ACI ($\gamma = 0.05$) over four base models, with coverage above each bar and the width change in bold. (b) Width change of Transport+ACI against the over-coverage of the static split in the same cell, for the eight base models of the cross-model analysis (blue) and the held-out early-stopped hurdle (orange); line and band: least-squares fit on the 40 cells with its 95% day-block bootstrap band.

with the same step has a residual of +4.0%. With the small steps selected by rolling origin the layers move little and almost entirely mechanically (R^2 of 0.98 and 0.99), and the residual is within a fraction of a per cent of zero (+0.8% for Transport+ACI and −0.1% for ACI). The slope of the cross-cell relationship is therefore mechanical. On average over the cells, adaptation in time contributes at most a fraction of a per cent of width beyond the best constant level chosen after the fact, and with the step of Algorithm 1 it costs several per cent.

What is not mechanical is where the over-coverage comes from. Within a window all base models face the same shift, and there the width change still follows the over-coverage (Table 5): in four of the five windows the correlation across the nine base models lies between −0.99 and −0.80; in the first semester of 2025, when the static splits of all base models are close to nominal and their over-coverage spans only 1.4 points, it is −0.18. In the transition window the static over-coverage ranges from −8.4 to +5.3 percentage points across base models, and in a two-way analysis of variance of the over-coverage over all cells the base model accounts for 54% and the window for 18%. Figure 8 shows the mechanism for the three main base models: the PIT histogram of the reference hurdle loses mass in the upper tail between the calibration and the transition windows, so that the calibrated quantile $\hat{q}$ over-covers, whereas that of the quantile model barely moves.

The cells share days, base models and windows, and Table 5 reports the slope under estimators that treat that dependence differently. Fixed effects for windows, for base models or for both, and crossed random effects, give slopes between −6.14 and −4.50. Over the 45 cells, the day-block bootstrap interval, which recomputes every cell on each resample of days, is [−5.66, −3.36]. The wild cluster bootstrap rejects a zero slope clearly when clustering by base model ($p = 0.0006$) and only narrowly when clustering by window ($p = 0.048$), with five clusters. These intervals describe uncertainty over the days of this record; with two model families they cannot describe uncertainty over model families, and we do not extend the attribution beyond the families tested.

5.5. Controlled simulation

Table 6 and Figure 9 report the simulation of Section 4.10. Three results matter. First, a correctly specified base model keeps the static split close to nominal coverage under every scenario, including a tripling of magnitudes: its mean over-coverage after the shift is +0.6 points in the level scenario and +0.1 points when level and dispersion

Table 4: How much of the adaptive width change is mechanical. For each adaptive configuration, the cross-cell regression of the width change on the static over-coverage (observed), the same regression for the constant-threshold oracle that attains the adaptive layer's coverage (mechanical), the share of the variance of the observed change explained by the mechanical one, and the mean residual. 40 cells: eight base models by five windows; 45 cells: the same plus the early-stopped hurdle. Intervals are 95% day-block bootstrap intervals, recomputing every cell on each of 2,000 replicates.

Configuration	Cells	Observed slope (%/pp)	Mechanical slope (%/pp)	R^2	Mean residual (%)	r(res., OC)
Transport+ACI, $\gamma = 0.05$	40	-4.89 [-5.60, -3.20]	-4.95 [-5.67, -3.34]	0.87	+7.0 [+5.3, +8.9]	+0.03
	45	-4.98 [-5.66, -3.36]	-4.80 [-5.53, -3.21]	0.84	+6.9 [+5.3, +8.6]	-0.07
Transport+ACI, $\gamma = 0.05$, $\alpha_{\min} = 0.005$	40	-6.28 [-7.26, -4.18]	-5.15 [-5.86, -3.50]	0.28	+11.7 [+8.6, +15.1]	-0.15
	45	-6.35 [-7.33, -4.31]	-4.98 [-5.69, -3.33]	0.30	+11.2 [+8.1, +14.5]	-0.19
ACI, $\gamma = 0.05$, $\alpha_{\min} = 0.005$	40	-4.98 [-5.52, -3.52]	-4.13 [-4.73, -2.82]	0.78	+4.3 [+2.6, +6.2]	-0.33
	45	-4.89 [-5.45, -3.45]	-3.98 [-4.59, -2.71]	0.79	+4.0 [+2.4, +5.9]	-0.36
Transport+ACI, selected	40	-2.52 [-3.36, -1.32]	-2.17 [-3.05, -0.99]	0.99	+1.1 [+0.4, +1.8]	-0.47
	45	-2.27 [-3.20, -1.07]	-1.88 [-2.85, -0.72]	0.98	+0.8 [+0.2, +1.5]	-0.50
ACI, selected	40	-1.38 [-1.71, -0.85]	-1.26 [-1.58, -0.74]	0.99	-0.1 [-0.3, +0.1]	-0.49
	45	-1.30 [-1.64, -0.76]	-1.17 [-1.51, -0.66]	0.99	-0.1 [-0.3, -0.0]	-0.49

Table 5: Inference on the cross-cell relationship under dependence, Transport+ACI with $\gamma = 0.05$. Slope of the width change (%) on the static over-coverage (pp). Rows differ in what they hold fixed and in how they treat the dependence between cells that share days, base models and windows.

Estimator	40 cells	45 cells
Pooled OLS	-4.89 (0.44)	-4.98 (0.41)
Window fixed effects	-5.62 (0.33)	-5.64 (0.34)
Base-model fixed effects	-4.23 (0.68)	-4.50 (0.62)
Two-way fixed effects	-6.00 (0.55)	-6.14 (0.59)
Crossed random effects (REML)	-5.58 (0.35)	-5.56 (0.35)
Day-block bootstrap, 95% interval	[-5.60, -3.20]	[-5.66, -3.36]
Wild cluster bootstrap by base model, p	0.0027 (8 cl.)	0.0006 (9 cl.)
Wild cluster bootstrap by window, p	0.0560 (5 cl.)	0.0484 (5 cl.)
Within-window correlation, range over 5 windows	[-0.99, -0.58]	[-0.99, -0.18]
Family × window means (10 cells), r	-0.84	-0.89
Variance of the static over-coverage explained (two-way ANOVA)		
base model	55%	54%
window	17%	18%

Standard errors in parentheses are the usual OLS or REML ones and treat cells as independent; they are shown for comparison with the bootstrap and wild cluster results. Wild cluster bootstrap: restricted, six-point Webb weights, 9,999 replicates.

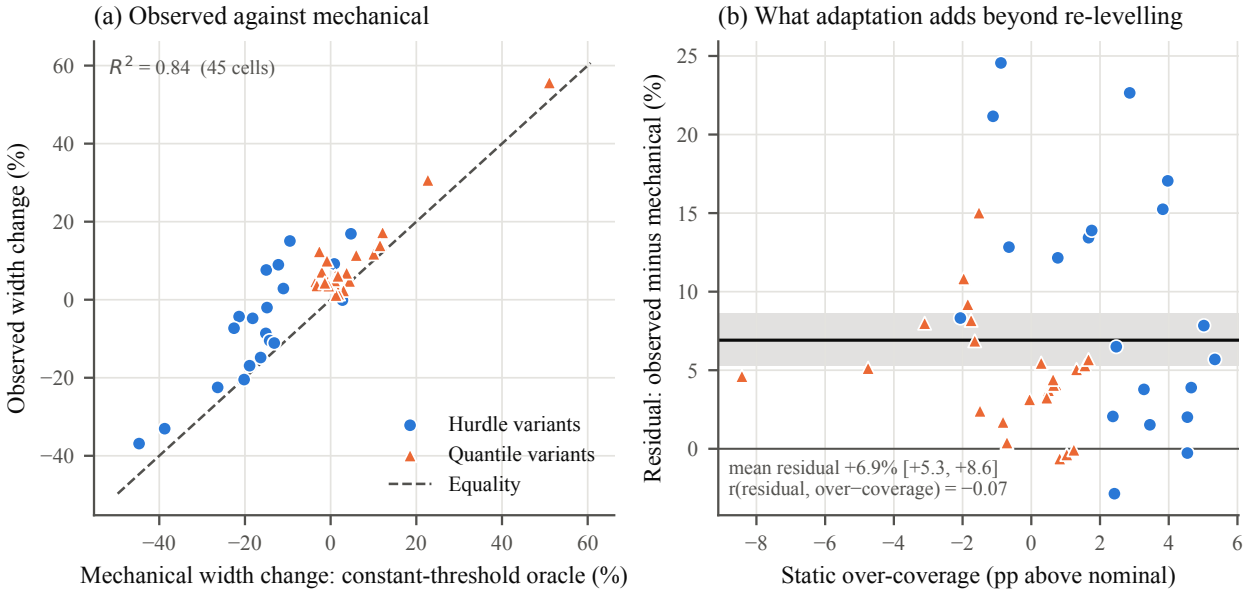


Figure 7: Decomposition of the width change of Transport+ACI ($\gamma = 0.05$) over the 45 base-model-by-window cells. (a) Observed width change against the width change of the constant-threshold oracle with the same coverage. (b) Residual against the static over-coverage; line and band: mean residual and its 95% day-block bootstrap interval.

Table 6: Controlled simulation, 50 replicates per scenario. Left: static over-coverage (pp) averaged over the four post-shift windows and the replicates, for four of the eight base models (Appendix F). Right: cross-cell regression of the Transport+ACI width change on the static over-coverage within each replicate (40 cells), median and 2.5–97.5 percentiles over replicates; mechanical slope from the constant-threshold oracle; share of the variance of the observed change explained by the mechanical one; mean residual.

	Static over-coverage (pp)				Cross-cell regression, Transport+ACI			
Scenario	M1	M2	M4	M7	Slope	Mech. slope	R^2 obs.~mech.	Residual (%)
S0, no shift	+0.1	+0.1	+0.1	+0.1	-7.18 [-17.09, -1.15]	-5.03 [-12.32, 0.20]	0.72 [0.03, 0.96]	10.7 [6.1, 35.9]
S1, level ×3	+0.6	+0.6	+0.6	-0.7	-7.88 [-16.39, -2.73]	-5.21 [-9.68, -0.18]	0.78 [0.10, 0.98]	12.3 [7.2, 32.7]
S2, level ×3 and dispersion ×0.6	+0.1	+0.1	+4.7	-1.6	-6.21 [-8.73, -4.14]	-5.22 [-7.15, -3.85]	0.90 [0.66, 0.99]	10.7 [4.9, 18.6]
S3, dispersion ×0.6	+0.0	+0.0	+4.7	+0.0	-5.59 [-8.46, -3.48]	-5.13 [-7.53, -3.18]	0.89 [0.66, 0.99]	10.8 [5.3, 17.3]

M1: correctly specified oracle. M2: oracle with dispersion inflated by 30% in every period. M4: dispersion fixed at its calibration value. M7: level lagged by 30 days. In S0 there is no shift and the regression has no systematic variation to explain.

change together, so an adaptive layer has nothing to recover. Second, a constant over-dispersion of 30 per cent, present in calibration and test alike, is absorbed by the conformal quantile: the over-coverage of that model is +0.1 points without a shift. Third, coverage drifts when the base model fails to follow the change: the constant-dispersion model over-covers by +4.7 points when the dispersion falls, and the lagged-level model under-covers during the ramp. In the three scenarios with a shift the cross-cell relationship appears in every replicate, and the constant-threshold oracle reproduces it: the median share of the variance of the observed width change explained by the mechanical one is 0.90 in the level-and-dispersion scenario, and the median residual is +10.7%, positive as in the real panel. The simulation also shows how much a single dataset can vary: with common daily shocks, the over-coverage of the correctly specified model in a single post-shift window ranges from −5.7 to +4.0 points across replicates (2.5th to 97.5th percentiles).

5.6. Model form, fitted capacity and early stopping

The comparison of base models holds features, rows, training cut-off, per-model hyperparameters and seed fixed, but not the number of fitted trees: the reference hurdle has 800, the reference quantile model 21,600. Table 7 separates the two. At a matched budget of about 800 trees the quantile model improves the CRPS over the whole test by only

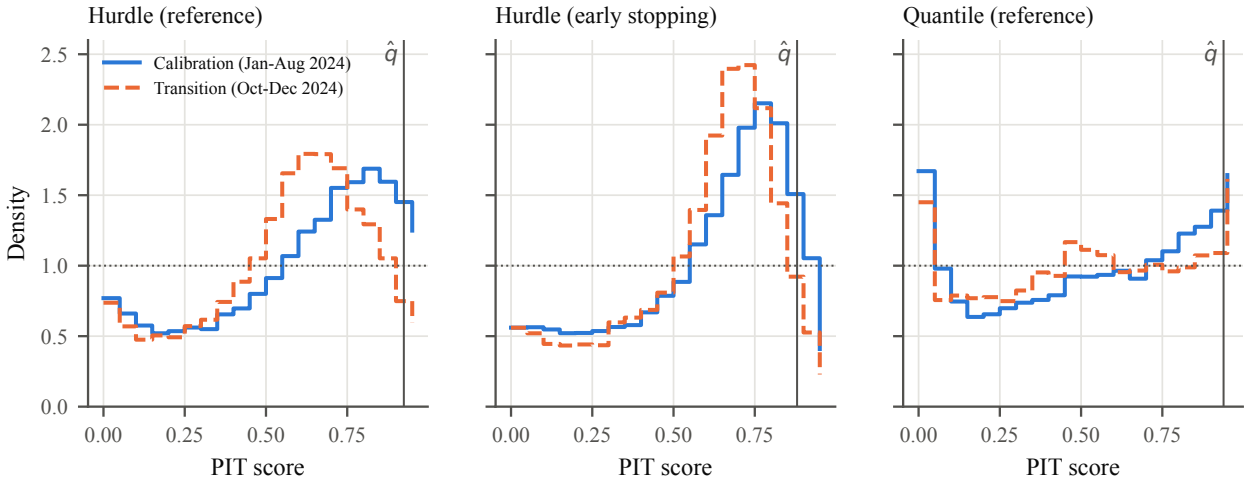


Figure 8: Histograms of the PIT score in the calibration window (January to August 2024) and in the transition window (October to December 2024) for the three main base models. The dotted line is the uniform density; the vertical line marks the static conformal quantile $\hat{q}$ at the 90% level.

Table 7: Model form and fitted capacity. Every model shares features, rows, training cut-off, per-model hyperparameters and seed; only the number of trees and the model form change. CRPS of the base predictive (MWh); change over the whole test relative to the reference hurdle; static-split coverage and mean width over the whole test.

		CRPS			Static split	
Base model	Trees	Transition	Whole test	vs. ref.	Coverage	Width
Hurdle, 2×400	800	133.5	89.5	+0.0%	92.5	678
Hurdle, early stopping (59+46)	105	124.0	84.4	-5.7%	92.0	409
Hurdle, 2×2000	4,000	141.1	95.5	+6.6%	92.1	970
Quantile GBM, 54×15	810	131.2	87.4	-2.4%	87.0	247
Quantile GBM, 54×50	2,700	100.6	71.4	-20.3%	89.4	263
Quantile GBM, 54×100	5,400	94.0	68.3	-23.7%	90.1	277
Quantile GBM, 54×200	10,800	92.8	67.8	-24.2%	90.2	285
Quantile GBM, 54×400	21,600	93.0	68.0	-24.1%	90.2	288

2.4%; the advantage grows with capacity and saturates near 5,400 trees, where it reaches 23.7%. The hurdle does not convert capacity into accuracy: with 4,000 trees its CRPS worsens by 6.6%, and early stopping, which selects 105 trees, improves it by 5.7%. The early-stopped hurdle remains 24% behind the quantile model. On this panel, and between these two families, the quantile form converts capacity into predictive accuracy and the two-part lognormal form saturates at about a hundred trees; we make no claim about the two model classes in general.

5.7. Which components of the transport-based scheme earn their cost

Table 8 attributes the interval score of Transport+ACI to its components, for the three main base models. Online adaptation helps where the static split over-covers: it improves the whole-test interval score by 95.9 MWh on the reference hurdle and by 19.8 MWh on the early-stopped hurdle, at a runtime of a fraction of a second, but it costs 8.5 MWh on the quantile model, whose static split is already at nominal coverage. Score transport without adaptation improves on the static split on the two hurdles by an amount similar to that of adaptation alone, so the two are substitutes. Once adaptation is present, adding the transport worsens the interval score on every base model, by +68.6, +30.3 and +11.5 MWh, each with an interval that excludes zero. The tail shrinkage changes the score by less than one per cent while accounting for 90% of the runtime of the scheme. The full scheme is not distinguishable from the static split on the two hurdles and is worse than it on the quantile model, by 19.1 MWh [+14.1, +24.2].

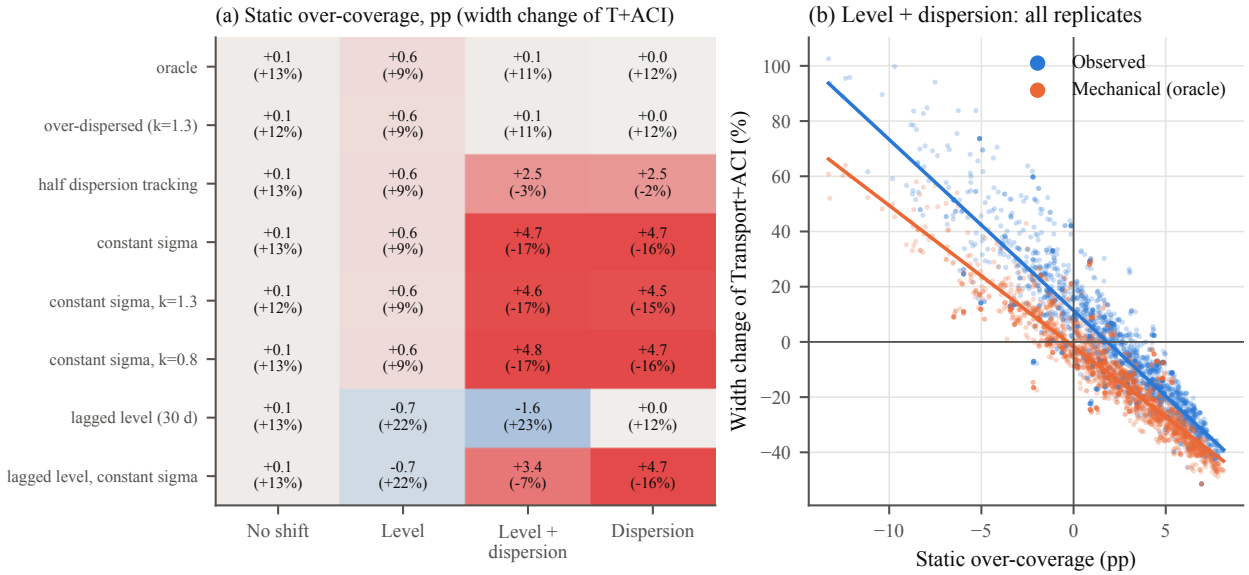


Figure 9: Controlled simulation. (a) Static over-coverage averaged over the post-shift windows and replicates, by base model and scenario, with the mean width change of Transport+ACI in parentheses. (b) Level-and-dispersion scenario, all cells of all replicates: observed width change of Transport+ACI (blue) and width change of the constant-threshold oracle (orange) against the static over-coverage, with least-squares lines.

Table 8: Marginal contribution of each component of the transport-based scheme to the mean one-sided interval score (MWh; negative is an improvement), with 95% day-block bootstrap intervals, for the three main base models. All adaptive steps use $\gamma = 0.05$ and no bound on α_t, as in Algorithm 1. Differences are computed on the plant-days where both limits are finite. Runtime is the additional wall-clock time of the component over the 638 test dates.

	Reference hurdle		Early-stopped hurdle		Reference quantile GBM		
Component	Transition	Whole test	Transition	Whole test	Transition	Whole test	Runtime (s)
Online adaptation (ACI over static)	-297.9†	-95.9†	-85.1†	-19.8†	+0.5	+8.5†	+0.22
Transport alone (over static)	-282.9†	-94.2†	-64.2†	-16.9†	+10.3†	+5.7†	+7.45
Transport added to ACI	+12.0	+68.6†	+12.3†	+30.3†	+14.1†	+11.5†	+0.46
Tail shrinkage added to transport	-0.6	-6.4†	-0.6†	-2.7†	-0.1	-0.3†	+6.36
Full scheme (over static)	-286.5†	-32.3	-73.4†	+9.4	+14.5†	+19.1†	+7.04

† the 95% interval excludes zero. Intervals for every cell are in the result files of the companion repository; runtime measured on the reference hurdle.

5.8. Operational evaluation

Table 9 reports coverage weighted by capacity and by curtailed energy, the share of curtailed energy left above the limit, and the expected cost at three cost ratios; Table 10 and Figure 10 report the behaviour on high-curtailment days.

Plant-day coverage overstates protection. Weighted by curtailed energy, the coverage of the methods built on the three main base models falls to between 63 and 84%, because limits are exceeded mostly on large days; weighting by installed capacity lowers it by one to three points. The transport-based variants on the reference hurdle leave between 13.3 and 14.2% of the curtailed energy above the limit, the highest shares among these methods, against 10.0% for the static split on the same base model and 9.8% for the static split on the quantile model. With every base model, Transport+ACI leaves more energy uncovered than the static split on the same model.

On high-curtailment days defined by the outcome, every method loses coverage relative to its average, and only methods that over-cover on average stay at or above the nominal level. On the 254 days of definition H1 the static split on the reference hurdle covers 91.9% and Transport+ACI 86.7%; the static split and CQR on the quantile model cover 86.7% and 85.9%, with limits less than half as wide as those of the reference hurdle's static split. Conditioning only on information available at the forecast origin (H3), the methods on the quantile model are close to nominal, at 91.0% for the static split and 90.4% for CQR, and Transport+ACI on the reference hurdle covers 89.6%. The shortfall

Table 9: Operational evaluation over the whole test period. Coverage by plant-day (date-clustered standard error), weighted by installed capacity and weighted by curtailed energy; share of curtailed energy above the limit; mean width (MWh); fraction of infinite limits. Cost: expected cost per plant-day in units of the allowance cost c_h, for a ratio $k = c_p/c_h$ of the penalty to the allowance cost, with the limit calibrated at the decision-optimal level $1 - 1/k$; at $k = 10$ it is the one-sided interval score at 90%. Costs over finite limits; an asterisk marks methods with infinite limits.

	Coverage and exposure (90% limits)						Cost per plant-day		
Method	Plant-day	Capacity	Energy	Uncovered	Width	% inf.	$k = 10$	$k = 20$	$k = 100$
Reference hurdle (2×400 trees)									
Static split (PIT)	92.5	90.9	80.2	10.0%	678	0.0	826	1320	3629
ACI (selected γ)	90.9	89.1	76.8	11.7%	590	0.0	763	1221	4225
Transport+ACI (selected)	90.0	88.0	74.0	13.3%	564	0.0	761	1302	4621
Transport+ACI ($\gamma = 0.05$)	90.1	88.0	73.0	13.8%	596	5.9	813*	–	–
One-sided CQR	91.2	88.7	74.8	12.4%	532	0.0	714	1181	3426
Early-stopped hurdle (105 trees)									
Static split (PIT)	92.0	90.7	78.8	8.4%	409	0.0	534	743	1693
ACI (selected γ)	90.8	89.2	75.9	9.7%	377	0.0	520	721	1748
Transport+ACI (selected)	89.9	88.1	73.1	10.9%	367	0.0	527	741	1769
Transport+ACI ($\gamma = 0.05$)	90.0	88.1	72.0	11.5%	377	3.3	552*	–	–
One-sided CQR	94.2	93.3	84.3	6.1%	501	0.0	590	1055	3634
Reference quantile GBM (54×400)									
Static split (PIT)	90.2	88.0	66.1	9.8%	288	0.0	433	539	929
ACI (selected γ)	90.3	88.1	66.2	9.8%	289	0.0	433	541	945*
Transport+ACI (selected)	90.1	87.9	65.6	10.1%	294	0.0	444	556	896*
Transport+ACI ($\gamma = 0.05$)	90.2	88.0	66.7	10.0%	305	4.4	459*	–	–
One-sided CQR	89.8	86.7	63.0	11.0%	276	0.0	438	546	927
Benchmarks without a base model									
Empirical quantile per plant, 365 days	87.2	87.0	61.9	11.5%	317	0.0	486	583	873
Rolling daysistribution per plant, 60 days	86.5	86.3	64.9	10.2%	282	0.0	433	531	912
Zero-inflated lognormal per plant	82.4	81.5	55.5	20.4%	338	0.0	639	942	3011

on outcome-defined days is the forecaster's dilemma [44]: a forecaster that is calibrated on average under-covers on the days that turned out extreme, unless its limits are wide everywhere.

At every cost ratio, the methods built on the quantile model are the cheapest among those that are close to nominal on average, and among them the calibration layer changes the cost by at most 5%. Transport+ACI on the reference hurdle costs 813 at $k = 10$ against 433 for the static split on the quantile model, and the selected Transport+ACI on the reference hurdle costs 1302 and 4621 at $k = 20$ and $k = 100$, against 539 and 929. In the tail the layers on the quantile model differ slightly: at $k = 100$ the selected Transport+ACI, whose step is small, is the cheapest of them (896, 3.5% below the static split) and the only one to reach 99% coverage. Two per-plant empirical quantiles that use no base model, over the last 60 days and over the last year, cost about as much as the methods on the quantile model, and the yearly one is the cheapest of all methods at $k = 100$ (873). Both under-cover at every level, at 86.5, 91.1 and 95.5% for the 60-day quantile and 87.2, 92.4 and 97.2% for the yearly one at the 90, 95 and 99% levels, which an owner who needs the stated level cannot accept.

5.9. Panel dependence and sensitivity

The intraclass correlation of the calibration scores by date is 0.238 (Section 4.3). Calibrating on one plant per date, which makes the scores approximately independent, moves the conformal quantile of the reference hurdle from 0.9242 to 0.9263 with a replicate standard deviation of 0.0112: the pooled construction misstates the precision of the guarantee, not its level. Mondrian calibration by technology or by capacity tercile changes little, and by region or by plant it worsens the worst regional deviation from 4.8 to 7.2 points, because some groups have too few calibration days for a stable quantile (Table G.20 in Appendix G). Among the main methods, the worst deviation across technologies, capacity terciles and regions ranges from 3.1 to 6.2 points (Table 10); the regional axis is the hardest for every method.

Appendix E varies every parameter of the calibration layer and of the evaluation. The step γ matters most: with $\alpha_{\min} = 0.005$, coverage stays within half a point of nominal for $\gamma \leq 0.05$ and falls to between 88 and 89% at $\gamma = 0.20$, and without a bound steps of 0.05 and above produce infinite limits on several per cent of plant-days. The length of the recent window comes next: a 30-day window raises the interval score of Transport+ACI on the early-stopped hurdle by 7%. The shrinkage scale, the bootstrap size, the grid of the map, the refresh period and the minimum

Table 10: Coverage on high-curtailment days and conditional coverage, 90% limits. H1: system curtailment above the calibration-window 90th percentile; H2: top decile of test days; H3: trailing seven-day mean of system curtailment known at the forecast origin above the same threshold. H1 and H2 condition on the outcome, H3 only on information available when the forecast is issued. Coverage on each subset with 95% day-block bootstrap interval; energy-weighted coverage on the same subset. The last three columns give the worst absolute deviation from 90% across technologies, capacity terciles and regions (pp).

	H1, ex post (254 days)		H2, ex post (64 days)		H3, ex ante (260 days)		Worst deviation		
Method	Coverage	Energy	Coverage	Energy	Coverage	Energy	Techn.	Size	Region
Reference hurdle (2×400 trees)									
Static split (PIT)	91.9 [90.6, 93.1]	79.6	90.7 [88.0, 92.9]	77.8	94.2 [93.4, 94.8]	82.1	2.5	3.6	4.8
ACI (selected γ)	90.2 [88.7, 91.4]	76.1	88.8 [85.8, 91.2]	74.2	92.6 [91.7, 93.5]	78.7	1.0	2.0	3.7
Transport+ACI (selected)	88.0 [86.4, 89.5]	72.1	85.6 [81.8, 88.9]	69.4	90.0 [88.9, 91.1]	73.4	0.1	1.8	4.9
Transport+ACI ($\gamma = 0.05$)	86.7 [85.2, 88.1]	69.2	84.1 [80.2, 87.4]	66.6	89.6 [88.4, 90.7]	72.2	0.3	1.6	4.8
One-sided CQR	89.9 [88.5, 91.1]	74.0	88.2 [85.3, 90.6]	71.7	92.6 [91.8, 93.4]	77.2	1.9	3.7	5.9
Early-stopped hurdle (105 trees)									
Static split (PIT)	91.4 [90.3, 92.4]	78.9	89.9 [87.6, 91.9]	76.8	94.4 [93.8, 94.9]	83.3	2.2	3.2	4.6
ACI (selected γ)	90.1 [88.9, 91.2]	76.1	88.4 [85.9, 90.6]	73.9	93.3 [92.6, 93.9]	80.8	0.9	2.0	3.7
Transport+ACI (selected)	88.0 [86.7, 89.3]	72.1	86.2 [83.0, 89.1]	70.4	90.9 [90.1, 91.8]	75.7	0.1	1.4	3.4
Transport+ACI ($\gamma = 0.05$)	86.5 [85.3, 87.8]	68.5	83.1 [79.6, 86.2]	64.8	89.9 [88.9, 90.9]	72.9	0.1	1.5	3.1
One-sided CQR	93.9 [92.9, 94.8]	84.5	92.8 [90.8, 94.4]	82.6	96.1 [95.7, 96.6]	88.1	4.4	5.0	6.0
Reference quantile GBM (54×400)									
Static split (PIT)	86.7 [85.3, 88.0]	62.5	82.2 [79.3, 84.9]	55.3	91.0 [90.2, 91.8]	69.8	1.8	2.4	5.9
ACI (selected γ)	86.6 [85.1, 87.9]	62.4	82.1 [79.0, 84.9]	55.5	90.9 [90.1, 91.7]	69.6	1.9	2.4	6.0
Transport+ACI (selected)	85.5 [84.1, 86.9]	60.5	80.4 [76.8, 83.5]	53.1	89.9 [89.0, 90.8]	67.6	1.7	2.5	6.1
Transport+ACI ($\gamma = 0.05$)	85.8 [84.2, 87.3]	61.6	81.5 [77.7, 85.0]	55.8	90.5 [89.5, 91.5]	69.2	1.9	2.3	6.1
One-sided CQR	85.9 [84.5, 87.1]	59.2	81.0 [78.2, 83.5]	51.4	90.4 [89.6, 91.2]	66.5	1.8	4.3	6.2

number of recent scores move it by at most 2.5%. On the quantile model the bound does not remove every infinite limit of Transport+ACI, because the transported pool can exceed the highest estimated quantile level, 0.999. The conclusions are unchanged with calibration windows of four and six months, with quantile grids of 27 and 14 levels, with other thresholds for high-curtailment days, and across ten seeds of the conformal layer, whose standard deviation of coverage is at most 0.04 points. Across depths, learning rates and training seeds of the early-stopped hurdle, its static split has an interval score between 517 and 558 MWh, always below Transport+ACI on the same model and far below the reference hurdle. The results for the transition window do not depend on where its boundaries are drawn (Table G.18).

6. Discussion

6.1. What the evaluation implies for practice

The width that an adaptive conformal layer saves against a static split is not a measure of how much distribution shift the layer absorbed. To first order it is the conversion of the static split's coverage error into width, which any layer that targets nominal coverage performs: the constant-threshold oracle, which adapts nothing in time, reproduces 84% of the variation of the observed change across cells, in the real panel and, with known ground truth, in the simulation. The static split's coverage error in turn measures how far the calibration of the base predictive drifted between the calibration and the evaluation window. With the window held fixed, so that every base model faces the same shift, that drift spans 13.8 percentage points across base models in the transition window, and the base model explains 54% of its variance against 18% for the window. The simulation shows the same structure: a correctly specified predictive keeps the static split near nominal coverage under a tripling of magnitudes, a dispersion that is wrong by the same factor in every period is absorbed by the conformal quantile, and coverage drifts only when the base model fails to follow a change in the data.

These results suggest three checks before a gain is attributed to an adaptive calibration layer: report the coverage of the static split in the evaluation window; report the width of the constant-threshold oracle at the adaptive layer's coverage, so that the residual, which is what adaptation adds, is visible; and repeat the comparison on a second base model. All three are cheap, because a well-designed layer touches the base model only through its CDF and quantile function.

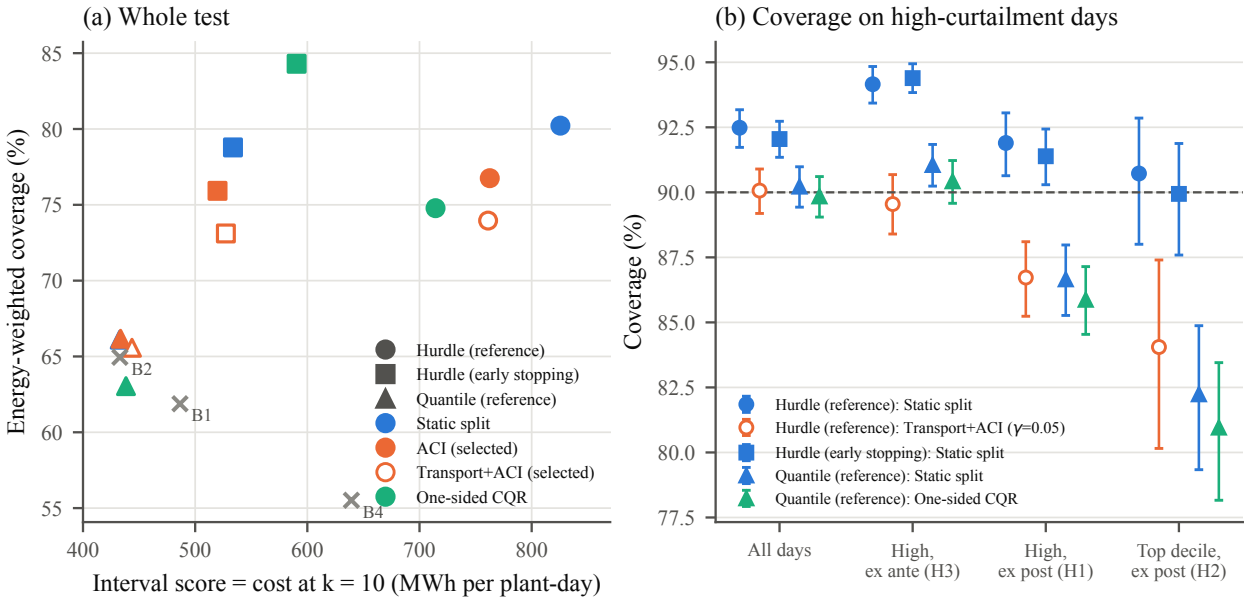


Figure 10: (a) Energy-weighted coverage against the 90% interval score, whole test; marker shape identifies the base model and colour the calibration layer, crosses are the benchmarks without a base model. (b) Coverage on all days and on high-curtailment days under the three definitions, with 95% day-block bootstrap intervals; dashed line: nominal 90%.

The ordering of effort follows. Replacing the reference hurdle by its early-stopped version, without touching the calibration layer, lowers the whole-test interval score of the static split by 35%, and replacing it by the quantile model lowers it by 48%. The best calibration layer on the reference hurdle lowers it by 13%. On a good base model the choice of layer matters little, and the ranking of layers itself depends on the base model: one-sided CQR is among the best layers on the quantile model and the worst on the early-stopped hurdle, because its correction is a single number of MWh pooled across plants of very different size (Section 5.3).

For an operator or asset owner working with public data at this horizon, the evidence supports the following practice.

1. Invest first in the base predictive: a flexible distributional model, regularised or early-stopped, whose PIT histogram is checked on the calibration window and on recent data.
2. Wrap it in a static conformal layer, a PIT split or a one-sided CQR, recalibrated periodically.
3. If drift is expected, add ACI with a step selected out of sample and a lower bound on the miscoverage level, $\alpha_{\min} = 0.005$, which removes infinite limits at a cost of at most half a point of coverage.
4. Check conditional coverage on regimes known when the forecast is issued, such as the trailing system curtailment, and report energy-weighted coverage next to plant-day coverage.
5. Do not use the transport-based scheme of Algorithm 1: once ACI is present the transport worsens the interval score, its tail shrinkage consumes most of the runtime for a negligible gain, and on outcome-defined high-curtailment days it leaves more curtailed energy uncovered than the static split.

6.2. The change in the record, and what we do not claim about it

The distributional change documented in Section 3.4 is a ramp of roughly eighteen months between mid-2023 and late 2024. The dataset cannot identify its causes, because it records curtailment and not the assets or network states that absorb it. Several mechanisms overlap: growth of solar capacity behind a saturated north–south corridor, reinforcements of that corridor, changes in northern mining demand, the entry of battery storage, and the saturation of curtailment once the corridor is full through the solar hours. Mohamad et al. [13] show that the effect of storage on curtailment depends on its position relative to the binding constraint, and Denholm and Mai [10] that it depends

on duration as much as on power; we have neither the network model nor the charge-discharge records that such an analysis requires.

A second phenomenon appears at the turn of 2024 to 2025: a break in the rate of growth. The year-over-year growth of the monthly mean log-magnitude falls to +0.09 per year for northern solar from January 2025 and to +0.01 at system level from December 2024, while the northern solar fleet keeps expanding by 8 per cent through 2025. Storage entry is one candidate for that flattening and is of the right order, but the supporting energy balance is an upper bound that assumes one full daily cycle, unit efficiency and that every charged megawatt-hour would otherwise have been curtailed, and the same exercise for January to May 2026 does not reproduce it. We record storage as one hypothesis among several and make no causal claim. The supplementary material lists the storage systems of the SEN with their commercial operation dates, so that the hypothesis can be tested by anyone with network and dispatch data.

The calibration layers respond to only one of these two kinds of change. A displacement of the level of the predictive law appears as a displacement of the score distribution and is what the transport map and the online update are built to absorb. A break in the rate of growth leaves no displacement to correct and is invisible to coverage, so reliability under shifts of level carries no guarantee under changes of trend.

6.3. The mid-2025 excursion

The coverage dip visible in Figure 5 is concentrated in June 2025, the largest deseasonalised monthly anomaly of the post-calibration record. The Wasserstein-1 distance between May to July 2025 and August to December 2025 is 1.334, larger than the 1.147 of the transition contrast. Its signature is the reverse of the 2024 window: magnitudes rise while the occurrence probability falls. It appears in solar and in wind and across four regions, which rules out a single plant or a residual erratum, and the operator's report for that month records no storage system entering operation. A one-month excursion of this kind is absorbed only slowly by a continuously adapting scheme and is the case a change-point trigger would catch.

6.4. Open directions

Four directions follow from the results. A change-point detector could trigger recalibration at the onset of a shift, which would address excursions like that of June 2025; the offline machinery of Section 3.4 is a starting point. A scale-invariant version of CQR, for instance with the nonconformity score normalised by a predicted scale or computed on the log scale, would remove the dependence of the correction on the mix of plant sizes and on changes in level. Localised conformal prediction [34], which weights the calibration set by proximity to the test point, is a more principled route to conditional validity than the Mondrian partitions of Appendix G. And the evaluation can be extended to portfolios of plants that share a transmission corridor and whose curtailment is jointly determined.

7. Limitations

1. *One system, one public data source.* All evidence comes from the SEN between 2022 and mid-2026. The mechanical part of the decomposition (Section 5.4) follows from how a coverage-targeting layer works and should hold elsewhere; its size, the coefficients of the cross-cell relationship and the ranking of methods should not be transported to other systems without checking.
2. *Two model families.* The nine base predictives are four variants of a lognormal hurdle and five capacity variants of a multi-quantile gradient-boosting model. The attribution of the coverage drift to the base model is established within this design; it is an empirical association for these families on this panel, and the intervals of Section 5.4 quantify sampling uncertainty over days, not uncertainty over model families.
3. *The oracle is a benchmark.* The constant-threshold oracle uses the outcomes of each window and cannot be deployed. It separates the part of a width change that requires no adaptation from the rest; it does not provide an alternative method.
4. *Dependence.* Exchangeability fails across the regime boundary and within it; the effective calibration sample is about 1,000 rows rather than 26,552 (Section 4.3). Block, group and plant-level calibration (Appendix G) quantify this but do not repair it. Differences of one to three points of coverage are within the date-clustered standard error and are not interpreted.

5. *High-curtailment days.* Coverage on days selected by the outcome is a description of exposure, not a test of calibration [44]; the ex-ante definition H3 is the legitimate test, and all definitions depend on a threshold, whose effect is reported in Appendix E.
6. *A stylised cost.* The cost model is linear in the allowance and in the exceedance, with constant unit costs, and ignores prices, contracts and the timing of settlement. It ranks methods for three cost ratios; it does not value them in money.
7. *A stylised simulation.* The simulated panel is generated from lognormal hurdles with common autoregressive shocks and linear ramps. It demonstrates the mechanism with known ground truth; its magnitudes are not estimates of the real ones.
8. *Selection on a short window.* Base-model predictions start in January 2024, so the rolling-origin selection uses four months of inner validation. That is enough to reject steps that produce infinite limits, but a thin basis for fine distinctions among the gentler ones.
9. *Point forecasting.* The comparison of Section 5.1 covers five model classes, twelve features, a seven-day horizon and mean absolute error. It says nothing about models with access to network topology, dispatch schedules, weather forecasts or storage state of charge, nor about criteria that weight large days.
10. *What the dataset contains.* The dataset records curtailment only, with no network state, dispatch decision, price or storage operation, so the causes of the distributional change are not identified (Section 6.2).

8. Conclusion

We have released an open, errata-corrected, plant-level dataset of renewable curtailment in Chile's National Electricity System, covering 53 continuous months and 300 plants and reconciling exactly with the operator's annual totals, and used it to evaluate conformal calibration of seven-day-ahead upper prediction limits under a documented change in the distribution of curtailment.

The main findings are four. First, the apparent benefit of an adaptive calibration layer depends on the base model: the scheme that narrows the reference hurdle's intervals by a third in the most divergent window leaves those of a multi-quantile model essentially unchanged. Second, to first order that benefit is the mechanical conversion of the static split's coverage error into width, which an oracle without any adaptation reproduces and a controlled simulation confirms; it measures the drift of the base predictive's calibration, not the size of the shift. Third, improving the base model dominates refining the calibration layer: early stopping alone narrows the hurdle's static limits by 40% at the same coverage, more than any calibration layer achieves on the reference hurdle. Fourth, on high-curtailment days defined by the outcome every method loses coverage and only those that over-cover on average stay at the nominal level, while on days identified from information available at the forecast origin the methods built on the quantile model stay close to nominal.

The transport-based adaptive pipeline is not the preferred operational method. On these data it under-covers on high-curtailment days (86.7% against 91.9% for the static split on the same base model), leaves more curtailed energy uncovered than the static split, and is dominated, at every cost ratio considered, by a static conformal wrapper (PIT split or one-sided CQR) around a multi-quantile predictive, which attains near-nominal coverage at less than half its width.

Three guarantees should be kept apart when such results are reported. The static split provides finite-sample marginal coverage under exchangeability, which this panel violates across the regime boundary and within it. The adaptive update provides long-run average coverage without distributional assumptions, and nothing at finite time. Empirical coverage in a dependent panel is a third thing: it is what we measure, and it varies across technologies, plant sizes, regions and high-curtailment days as reported. The dataset and the code that regenerates every table, figure and number are public.

CRediT authorship contribution statement

Pablo Reyes Cerda: Conceptualization, Data curation, Software, Validation, Investigation, Writing – original draft. **Kerven Cea Morales:** Methodology, Formal analysis, Validation, Writing – review and editing.

Declaration of competing interest

Pablo Reyes Cerda is the founder of Synapta SpA and of CurtailmentIQ, a commercial forecasting service that uses data sources related to those described in this article. Kerven Cea Morales declares no competing interest.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI-based assistant to support the drafting and language editing of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The dataset is openly available on Zenodo under a CC BY 4.0 licence (DOI: 10.5281/zenodo.21198816, which resolves to the latest version of the deposit). The processing pipeline, the modelling code and the scripts that regenerate every table, figure and number in this article are available in the companion repository at `https://github.com/Pablosebastian-reyes/curtailmentiq-research`. The supplementary material contains a registry of the energy storage systems of the SEN with their commercial operation dates and sources, compiled from the public reports of the system operator, the Ministry of Energy and the National Energy Commission; it is not part of the released dataset.

Appendix A. Inversion of the PIT limit

Fix x and write $p = p(x)$, $\mu = \mu(x)$. The mixture CDF is $F_x(y) = 0$ for $y < 0$, $F_x(0) = 1 - p$, and $F_x(y) = (1 - p) + p\,\Phi((\log y - \mu)/\sigma)$ for $y > 0$, which is continuous and strictly increasing on $(0, \infty)$ from $1 - p$ to 1. Given a level $\hat{q} \in [0, 1]$ we seek the smallest $u \geq 0$ with $F_x(u) \geq \hat{q}$. If $\hat{q} \leq 1 - p$ the atom already reaches the level, so $u = 0$. If $\hat{q} \geq 1$ no finite u attains it and $u = +\infty$. For $1 - p < \hat{q} < 1$, strict monotonicity gives a unique solution of $(1 - p) + p\,\Phi((\log u - \mu)/\sigma) = \hat{q}$, hence $\log u = \mu + \sigma\,\Phi^{-1}((\hat{q} - (1 - p))/p)$, which is (3). The boundary cases are the limits of the clipped argument, since $\Phi^{-1}(0^+) = -\infty$ gives $U = 0$ and $\Phi^{-1}(1^-) = +\infty$ gives $U = +\infty$. □

For the quantile model the same object is obtained without a closed form. Let $\tau_1 < \cdots < \tau_K$ be the estimated levels and $\hat{q}_{\tau_1}(x) \leq \cdots \leq \hat{q}_{\tau_K}(x)$ the rearranged quantiles, and augment the grid with the node $(0, 0)$. Then F_x is the piecewise-linear interpolant of the augmented pairs, its atom is $F_x(0) = \max\{\tau_k : \hat{q}_{\tau_k}(x) = 0\}$, and $F_x^{-1}(\hat{q})$ is the piecewise-linear interpolant in the other direction, equal to 0 for $\hat{q} \leq F_x(0)$ and to $+\infty$ for $\hat{q} > \tau_K$.

For both families, and for $y > 0$, $y \leq F_x^{-1}(q)$ if and only if $F_x(y) \leq q$. The coverage of a constant threshold q over a set of plant-days is therefore the fraction of zero outcomes plus the fraction of positive outcomes with $F_x(y) \leq q$, which is how the oracle of Section 4.9 is computed exactly.

Appendix B. Coverage under score transport

Static split. If the calibration scores $s_1, \ldots, s_n$ and the test score s_{n+1} are exchangeable, then with $\hat{q} = s_{(k)}$, $k = \lceil (n + 1)(1 - \alpha) \rceil$, one has $\mathbb{P}(s_{n+1} \leq \hat{q}) \geq k/(n + 1) \geq 1 - \alpha$, and since F_x is monotone, $s_{n+1} \leq \hat{q}$ if and only if $Y \leq U(x)$; hence $\mathbb{P}(Y \leq U(x)) \geq 1 - \alpha$. The statement is marginal and exact only under exchangeability; with an intraclass correlation by date of 0.238 and 108.8 plants per day, the effective calibration sample is about 1,000 rather than 26,552, and the guarantee should be read at that size.

Transported scores. Let P_{old} and P_{new} be the laws of the score in the old and new regimes and $T = F_{\text{new}}^{-1} \circ F_{\text{old}}$ the monotone transport map. Under an oracle map, $T(s) \sim P_{\text{new}}$ for $s \sim P_{\text{old}}$, so the transported calibration scores are exchangeable with the test score and the static guarantee transfers. With an estimated map $\hat{T}$ the coverage gap is bounded by the total-variation distance $d_{\text{TV}}(\hat{T}_{\#}P_{\text{old}}, P_{\text{new}})$ through the framework of Barber et al. [32]. The bound cannot be computed from data, so validity is obtained by feeding the transported scores into the online update of Gibbs and Candès [2], whose long-run coverage holds for any sequence; the transport contributes sharpness and the update contributes coverage. Section 5.7 shows that on these data the sharpness contribution is negative once the update is present.

Appendix C. Hyperparameters, constants and seeds

Table C.11 lists the constants of the pipeline and the values selected by rolling origin for each base model.

Table C.11: Hyperparameters, constants, selected values and seeds.

Item	Value	Detail
Base models		
Forecast horizon	7 days	direct; every dynamic feature lags the target by at least 7 days
Last training target	2023-12-31	no model sees the evaluation period
Trees per model / learning rate / depth	400 / 0.05 / 6	fixed a priori; reference models
Min. child weight / subsample / colsample	5 / 0.9 / 0.9	all models
Features	12	lags 7, 8, 14, 21 d; 7- and 30-d means; occurrence; day of week and year; log capacity; region; technology
Hurdle dispersion	out of fold, 5 folds	$\hat{\sigma}$ = 1.70 (reference), 1.83 (early stopping)
Early stopping of the hurdle	ceiling 4,000, patience 50	validation Jul–Dec 2023; refit on full training period: 59 + 46 trees
Quantile grid	54 levels, 0.005–0.999	0.02–0.98 by 0.02, plus 0.005, 0.01, 0.99, 0.995, 0.999; rearranged
Capacity ladder	15, 50, 100, 200, 400	trees per quantile level
Calibration layers		
Nominal level	$1 - \alpha = 0.90$	one-sided upper limit $[0, U]$; tail levels 0.95 and 0.99
Score	PIT, $s = F_x(y)$	atom at zero randomised
Calibration window	2024-01-01 to 2024-08-31	244 days, 26,552 plant-days
Horizon embargo	7 days	online windows end at $t-7$; error feedback delayed 7 steps
Sliding window	60 days, refreshed every 7	
Transport window / refresh	60 days / 7 days	at least 100 recent scores per refresh
Transport quantile grid	$g = \min(100, \max(20, m/2))$	m recent scores; two linear interpolations
Bootstrap of the map / shrinkage scale	$B = 150$ / $\tau = 0.05$	$\lambda(u) = [1 + (\mathrm{sd}_{\mathrm{boot}}(u)/\tau)^2]^{-1}$
ACI bound	none, or $\alpha_{\min} = 0.005$	both reported
One-sided CQR	$U = \max\{q_{0.9}(x) + \hat{Q}_{0.9}(y - q_{0.9}(x)), 0\}$	on each base model's own quantile
Selection by rolling origin		
Inner calibration / validation	Jan–Apr 2024 / May–Aug 2024	same refresh and embargo as the test
Grids	step γ: 0.005, 0.01, 0.02, 0.05, 0.10, 0.20	windows: 30, 45, 60, 90, 120 days
Criterion	mean one-sided interval score	$+\infty$ if any limit is infinite
Selected, reference hurdle	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 120 d
Selected, early-stopped hurdle	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 120 d
Selected, hurdle with $\sigma(x)$	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 120 d
Selected, hurdle 2×2000	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 120 d

Table C.11, continued.

Item	Value	Detail
Selected, quantile GBM 54×15	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 45 d
Selected, quantile GBM 54×50	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 45 d
Selected, quantile GBM 54×100	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 45 d
Selected, quantile GBM 54×200	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 45 d
Selected, reference quantile GBM	ACI $\gamma = 0.005$	Transport+ACI $\gamma = 0.005$, window 45 d
Evaluation		
Primary metric	$U + \alpha^{-1}\max(y - U, 0)$	proper for the $1 - \alpha$ quantile; $+\infty$ if U is
Coverage standard error	clustered by date	
Bootstrap of differences	2,000 replicates of whole days	95% percentile interval
CRPS / twCRPS	300 samples per plant-day	twCRPS threshold: plant's calibration-window 90th percentile
Seeds		
Base-model training	42	XGBoost hist, 4 threads
PIT atom and map bootstrap	20260720	one generator, consumed in a fixed order
CRPS sampling	11	
Day-block bootstrap and wild cluster bootstrap	20260901	
Simulation	20261103	seed per scenario and replicate

Appendix D. Reproducibility

Every table, figure and number in the text is regenerated by versioned scripts from the frozen dataset; the numbers quoted in the text are read from the result files by a script rather than transcribed. The environment is Python 3.12.3 with pandas 3.0.3, numpy 2.4.6, scipy 1.17.1, scikit-learn 1.9.0, xgboost 3.2.0, ruptures 1.1.10, statsmodels 0.15.0 and matplotlib 3.11.0. The seeds are 42 for base-model training, 20260720 for the randomisation of the PIT atom and the bootstrap of the transport map, 11 for the CRPS sampling, 20260901 for the day-block bootstraps, the permutation nulls and the wild cluster bootstrap, and 20261103 for the simulation. The order in which the conformal generator is consumed is fixed and documented, so that the reference hurdle reproduces its published results bit for bit. All nine base models were refitted from the training data with the published code: the reference models reproduce the stored predictions to a relative difference below 10^{-5}, which is the rounding of the stored files, and the early-stopped hurdle reproduces its tree counts and dispersion exactly. The command sequence and a log of every run are in the companion repository.

Appendix E. Sensitivity analysis

All analyses vary one parameter at a time around a reference configuration, Transport+ACI with $\gamma = 0.05$, a 60-day window, weekly refresh, $B = 150$, $\tau = 0.05$, a map grid of 100 levels, at least 100 recent scores and $\alpha_{\min} = 0.005$, and report the whole test period. The embargo is fixed at seven days by the forecast horizon and is not a free parameter. Table E.12 covers the calibration layer, Table E.13 the nominal level, Table E.14 the calibration window, the quantile grid and the threshold defining high-curtailment days, and Table E.15 the seeds and the base-model hyperparameters. The full γ×window grid for the reference hurdle is in Table G.19 of Appendix G.

Table E.12: Sensitivity of Transport+ACI to the parameters of the calibration layer, whole test, for the early-stopped hurdle and the reference quantile GBM. Coverage (%), mean width (MWh), fraction of infinite limits (%) and one-sided interval score over finite limits (MWh).

		Early-stopped hurdle				Reference quantile GBM			
Parameter	Value	Cov.	Width	% inf.	IS	Cov.	Width	% inf.	IS
Reference	ref	89.8	375	0.0	547	89.9	302	1.4	455
Shrinkage scale τ	0.01	89.8	375	0.0	546	89.9	302	0.9	454
Shrinkage scale τ	0.025	89.8	375	0.0	547	89.9	302	1.4	454
Shrinkage scale τ	0.1	89.8	375	0.0	547	89.9	301	1.5	454
Shrinkage scale τ	0.25	89.8	375	0.0	547	89.9	301	1.5	454
Shrinkage scale τ	none	89.8	375	0.0	547	89.9	301	1.5	454
Bootstrap size B	25	89.8	376	0.0	547	89.9	301	1.7	454
Bootstrap size B	50	89.8	375	0.0	547	89.9	300	1.8	454
Bootstrap size B	300	89.8	375	0.0	547	89.9	302	1.4	454
Map grid	25	89.8	378	0.0	550	89.9	303	0.6	454
Map grid	50	89.8	377	0.0	548	89.9	303	0.6	455
Map grid	200	89.8	374	0.0	546	89.9	303	0.9	455
Refresh period (days)	1	89.8	380	0.0	551	89.9	302	1.2	455
Refresh period (days)	14	89.7	366	0.0	540	89.9	302	1.1	455
Refresh period (days)	28	89.7	359	0.0	533	89.9	302	0.2	454
Min. recent scores	50	89.8	375	0.0	547	89.9	302	1.4	455
Min. recent scores	500	89.8	375	0.0	547	89.9	302	1.4	455
Min. recent scores	2000	89.8	375	0.0	547	89.9	302	1.4	455
Window (days)	30	89.8	415	0.0	584	89.7	308	1.8	465
Window (days)	45	89.7	389	0.0	561	89.8	305	1.8	459
Window (days)	90	89.7	372	0.0	545	89.9	295	1.9	451
Window (days)	120	89.8	364	0.0	534	90.0	298	1.5	451
Transport without ACI	gamma=0	89.8	350	0.0	517	90.0	288	0.0	438
Step γ	0.005	89.9	357	0.0	524	90.2	293	0.0	441
Step γ	0.01	90.0	362	0.0	531	90.2	295	0.0	443
Step γ	0.02	90.0	381	0.0	551	90.2	299	0.0	447
Step γ	0.1	89.5	409	0.0	578	89.5	293	3.7	450
Step γ	0.2	88.5	498	0.0	681	88.0	286	9.5	470
Bound $\alpha_{\min}$	none	90.0	379	3.3	554	90.2	304	4.4	459
Bound $\alpha_{\min}$	0.001	89.8	400	0.0	572	89.9	301	1.9	454
Bound $\alpha_{\min}$	0.01	89.7	365	0.0	537	89.9	302	0.5	454
Bound $\alpha_{\min}$	0.02	89.7	355	0.0	528	89.8	299	0.0	451
ACI, step γ	0.005	90.8	377	0.0	520	90.3	289	0.0	434
ACI, step γ	0.01	90.4	367	0.0	517	90.2	289	0.0	434
ACI, step γ	0.02	90.2	361	0.0	519	90.2	290	0.0	437
ACI, step γ	0.1	89.7	386	0.0	552	89.5	298	0.0	452
ACI, step γ	0.2	88.9	445	0.0	619	88.8	315	0.0	474
ACI, step γ	0.05	89.8	355	0.0	521	90.0	294	0.0	444

Table E.13: Sensitivity to the nominal level, whole test. Coverage (%) and one-sided interval score at the same level (MWh). Adaptive layers use $\gamma = 0.05$ and $\alpha_{\min} = \alpha/20$.

		Early-stopped hurdle		Ref. quantile GBM	
Level	Method	Cov.	IS	Cov.	IS
0.80	Static split	83.8	380	82.1	343
0.80	ACI	80.2	382	80.3	348
0.80	Transport+ACI	80.0	394	80.2	356
0.80	One-sided CQR	82.4	372	81.1	348
0.90	Static split	92.0	534	90.2	433
0.90	ACI	89.8	521	90.0	444
0.90	Transport+ACI	89.8	547	89.9	455*
0.90	One-sided CQR	94.2	590	89.8	438
0.95	Static split	96.0	743	94.3	539
0.95	ACI	94.8	701	94.7	555*
0.95	Transport+ACI	94.9	748	94.8	558*
0.95	One-sided CQR	98.0	1055	94.2	546
0.99	Static split	99.1	1693	98.4	929
0.99	ACI	99.0	1470	98.9	913*
0.99	Transport+ACI	98.9	1566	98.9	910*
0.99	One-sided CQR	99.8	3634	98.4	928

* the method returns infinite limits; the score is over finite limits.

Appendix F. Specification of the controlled simulation

Panel. $N = 100$ plants observed for 240 calibration days followed by 638 test days, split into windows of 30, 92, 181, 184 and 151 days, the lengths of the real windows; the second window plays the role of the transition window.

Data-generating process. For plant i and day t, occurrence is $Z_{it} \sim \text{Bernoulli}(\text{logit}^{-1}(a_i + \eta_t))$ and, when $Z_{it} = 1$, $\log Y_{it} = m_i + \ell_t + \xi_t + s_t\,\varepsilon_{it}$ with $\varepsilon_{it} \sim N(0,1)$. Plant effects are $a_i \sim N(1, 0.8^2)$ and $m_i \sim N(4, 0.8^2)$. The common shocks η_t and ξ_t are stationary first-order autoregressive processes with coefficient 0.7 and marginal standard deviations 0.8 and 0.5. They are the source of dependence across plants.

Scenarios. The level path ℓ_t and the idiosyncratic dispersion s_t are constant at 0 and 1 before the transition window, change linearly over it and stay constant afterwards. S0: no change. S1: ℓ rises by $\log 3$, a tripling of magnitudes. S2: as S1, and s falls from 1.0 to 0.6. S3: s falls from 1.0 to 0.6 with no change in level.

Base predictives. All are lognormal hurdles with occurrence probability equal to the true probability marginalised over η_t, which a forecaster seven days ahead cannot predict. They differ in location and dispersion. Writing $\sigma_t = (s_t^2 + 0.5^2)^{1/2}$ for the true marginal dispersion of $\log Y$ given occurrence and σ_{cal} for its calibration value, the predictive dispersion is $\kappa\,\sigma_{\text{cal}}^{1-\phi}\sigma_t^{\phi}$ and the location is $m_i + \ell_{t-L}$. M1, the oracle: $\phi = 1$, $\kappa = 1$, $L = 0$. M2: $\phi = 1$, $\kappa = 1.3$. M3: $\phi = 0.5$. M4, constant dispersion: $\phi = 0$. M5: $\phi = 0$, $\kappa = 1.3$. M6: $\phi = 0$, $\kappa = 0.8$. M7: $\phi = 1$, $L = 30$ days. M8: $\phi = 0$, $L = 30$ days.

Calibration and evaluation. The same code as for the real data: static PIT split on the calibration days; ACI and Transport+ACI with $\gamma = 0.05$, $\alpha_{\min} = 0.005$, a 60-day window, weekly refresh and the 7-day embargo; nominal level 0.90. For each model-window cell we compute the static over-coverage, the width change of each adaptive layer and the width change of the constant-threshold oracle at the adaptive layer's coverage. Fifty independent replicates per scenario. The seeds of the data, of the randomisation of the PIT atom and of the map bootstrap are fixed functions of the scenario, replicate and model indices, so every replicate can be regenerated on its own.

Appendix G. Additional tables

Tables G.16–G.20 complete the main text. Table G.16 gives the reference hurdle by window; Table G.17 the effect of bounding α_t; Table G.18 the sensitivity of the transition-window results to its boundaries; Table G.19 the full $\gamma\times$window grid on the test period, a display produced after the selection was closed; and Table G.20 the calibration schemes designed for panel dependence.

Table E.14: Sensitivity to the calibration window (all windows end on 31 August 2024), to the density of the quantile grid of the reference quantile GBM, and to the threshold that defines high-curtailment days (H1). Whole test; coverage (%) and one-sided interval score over finite limits (MWh).

Analysis	Value	Method	Coverage	IS
Calibration window	8 months (Jan-Aug) (early-stopped hurdle)	Static split	92.0	534
Calibration window	8 months (Jan-Aug) (early-stopped hurdle)	Transport+ACI	89.8	547
Calibration window	8 months (Jan-Aug) (early-stopped hurdle)	One-sided CQR	94.2	590
Calibration window	6 months (Mar-Aug) (early-stopped hurdle)	Static split	92.7	546
Calibration window	6 months (Mar-Aug) (early-stopped hurdle)	Transport+ACI	89.8	548
Calibration window	6 months (Mar-Aug) (early-stopped hurdle)	One-sided CQR	94.3	591
Calibration window	4 months (May-Aug) (early-stopped hurdle)	Static split	93.3	559
Calibration window	4 months (May-Aug) (early-stopped hurdle)	Transport+ACI	89.8	546
Calibration window	4 months (May-Aug) (early-stopped hurdle)	One-sided CQR	94.3	591
Calibration window	8 months (Jan-Aug) (quantile GBM)	Static split	90.2	433
Calibration window	8 months (Jan-Aug) (quantile GBM)	Transport+ACI	89.9	455
Calibration window	8 months (Jan-Aug) (quantile GBM)	One-sided CQR	89.8	438
Calibration window	6 months (Mar-Aug) (quantile GBM)	Static split	89.9	433
Calibration window	6 months (Mar-Aug) (quantile GBM)	Transport+ACI	89.9	455
Calibration window	6 months (Mar-Aug) (quantile GBM)	One-sided CQR	89.6	438
Calibration window	4 months (May-Aug) (quantile GBM)	Static split	89.6	433
Calibration window	4 months (May-Aug) (quantile GBM)	Transport+ACI	89.9	455
Calibration window	4 months (May-Aug) (quantile GBM)	One-sided CQR	89.2	439
Quantile grid	54 levels (quantile GBM)	Static split	90.2	433
Quantile grid	54 levels (quantile GBM)	Transport+ACI	89.9	454
Quantile grid	54 levels (quantile GBM)	One-sided CQR	89.8	438
Quantile grid	27 levels (quantile GBM)	Static split	90.2	433
Quantile grid	27 levels (quantile GBM)	Transport+ACI	89.9	455
Quantile grid	27 levels (quantile GBM)	One-sided CQR	89.8	438
Quantile grid	14 levels (quantile GBM)	Static split	90.2	433
Quantile grid	14 levels (quantile GBM)	Transport+ACI	89.9	455
Quantile grid	14 levels (quantile GBM)	One-sided CQR	89.8	438
High-day threshold	P75 (425 days), reference hurdle	Static split (PIT)	92.2	–
High-day threshold	P75 (425 days), reference hurdle	Transport+ACI (selected)	88.9	–
High-day threshold	P75 (425 days), reference hurdle	Transport+ACI ($\gamma = 0.05$)	88.1	–
High-day threshold	P75 (425 days), reference hurdle	One-sided CQR	90.5	–
High-day threshold	P90 (254 days), reference hurdle	Static split (PIT)	91.9	–
High-day threshold	P90 (254 days), reference hurdle	Transport+ACI (selected)	88.0	–
High-day threshold	P90 (254 days), reference hurdle	Transport+ACI ($\gamma = 0.05$)	86.7	–
High-day threshold	P90 (254 days), reference hurdle	One-sided CQR	89.9	–
High-day threshold	P95 (197 days), reference hurdle	Static split (PIT)	92.4	–
High-day threshold	P95 (197 days), reference hurdle	Transport+ACI (selected)	88.4	–
High-day threshold	P95 (197 days), reference hurdle	Transport+ACI ($\gamma = 0.05$)	87.0	–
High-day threshold	P95 (197 days), reference hurdle	One-sided CQR	90.4	–
High-day threshold	P75 (425 days), early-stopped hurdle	Static split (PIT)	91.6	–
High-day threshold	P75 (425 days), early-stopped hurdle	Transport+ACI (selected)	88.6	–
High-day threshold	P75 (425 days), early-stopped hurdle	Transport+ACI ($\gamma = 0.05$)	88.1	–
High-day threshold	P75 (425 days), early-stopped hurdle	One-sided CQR	93.9	–
High-day threshold	P90 (254 days), early-stopped hurdle	Static split (PIT)	91.4	–
High-day threshold	P90 (254 days), early-stopped hurdle	Transport+ACI (selected)	88.0	–
High-day threshold	P90 (254 days), early-stopped hurdle	Transport+ACI ($\gamma = 0.05$)	86.5	–
High-day threshold	P90 (254 days), early-stopped hurdle	One-sided CQR	93.9	–
High-day threshold	P95 (197 days), early-stopped hurdle	Static split (PIT)	92.2	–
High-day threshold	P95 (197 days), early-stopped hurdle	Transport+ACI (selected)	88.8	–
High-day threshold	P95 (197 days), early-stopped hurdle	Transport+ACI ($\gamma = 0.05$)	86.7	–
High-day threshold	P95 (197 days), early-stopped hurdle	One-sided CQR	94.6	–
High-day threshold	P75 (425 days), reference quantile GBM	Static split (PIT)	88.0	–
High-day threshold	P75 (425 days), reference quantile GBM	Transport+ACI (selected)	87.3	–
High-day threshold	P75 (425 days), reference quantile GBM	Transport+ACI ($\gamma = 0.05$)	87.7	–
High-day threshold	P75 (425 days), reference quantile GBM	One-sided CQR	87.4	–
High-day threshold	P90 (254 days), reference quantile GBM	Static split (PIT)	86.7	–
High-day threshold	P90 (254 days), reference quantile GBM	Transport+ACI (selected)	85.5	–
High-day threshold	P90 (254 days), reference quantile GBM	Transport+ACI ($\gamma = 0.05$)	85.8	–
High-day threshold	P90 (254 days), reference quantile GBM	One-sided CQR	85.9	–
High-day threshold	P95 (197 days), reference quantile GBM	Static split (PIT)	87.1	–
High-day threshold	P95 (197 days), reference quantile GBM	Transport+ACI (selected)	85.6	–
High-day threshold	P95 (197 days), reference quantile GBM	Transport+ACI ($\gamma = 0.05$)	85.8	–
High-day threshold	P95 (197 days), reference quantile GBM	One-sided CQR	86.1	–

Table E.15: Sensitivity to seeds and to base-model hyperparameters. Top: ten seeds of the conformal layer (PIT atom and map bootstrap), mean and standard deviation over seeds. Bottom: early-stopped hurdle refitted with other depths, learning rates and training seeds; trees selected by early stopping, CRPS over the whole test, and whole-test coverage and interval score of the static split and of Transport+ACI.

Analysis	Method	Coverage		IS	
		mean	sd	mean	sd
Conformal seeds, early-stopped hurdle	Static split	92.08	0.04	533.8	0.2
Conformal seeds, early-stopped hurdle	Transport+ACI	89.80	0.00	547.0	0.4
Conformal seeds, reference quantile GBM	Static split	90.20	0.00	432.7	0.0
Conformal seeds, reference quantile GBM	Transport+ACI	89.90	0.00	454.5	0.1
		Trees	CRPS	Static (cov./IS)	T+ACI (cov./IS)
Early-stopped hurdle	depth = 4	126	83.6	92.2 / 526	89.8 / 534
Early-stopped hurdle	depth = 8	92	86.1	91.9 / 558	89.8 / 580
Early-stopped hurdle	learning rate = 0.03	190	84.2	92.2 / 533	89.7 / 549
Early-stopped hurdle	learning rate = 0.1	56	84.5	92.3 / 546	89.8 / 554
Early-stopped hurdle	training seed = 1	112	84.2	92.1 / 541	89.7 / 556
Early-stopped hurdle	training seed = 2	115	84.2	92.2 / 526	89.7 / 547
Early-stopped hurdle	training seed = 3	99	84.2	92.0 / 523	89.7 / 541
Early-stopped hurdle	training seed = 4	111	84.0	92.2 / 541	89.7 / 546
Early-stopped hurdle	training seed = 5	119	84.2	92.3 / 528	89.7 / 546
Early-stopped hurdle	training seed = 6	97	84.0	92.2 / 526	89.7 / 535
Early-stopped hurdle	training seed = 7	108	84.5	91.9 / 532	89.7 / 553
Early-stopped hurdle	training seed = 8	129	84.5	92.2 / 557	89.7 / 570
Early-stopped hurdle	training seed = 9	92	84.7	91.8 / 517	89.7 / 536
Early-stopped hurdle	training seed = 10	96	84.5	92.2 / 525	89.7 / 539

Table G.16: Coverage and mean interval width of the conformal methods by period, on real data with the hurdle base model, with the 7-day horizon embargo. Each cell shows coverage in percent (clustered standard error by date in parentheses) over mean interval width in MWh. Nominal coverage is 90%. The base-model CRPS is a property of the predictive, not of the calibration layer, and is reported once per period. Superscripts give the fraction of infinite intervals where nonzero; Table 3 reports the interval score, which does not let those cases improve a method's score.

Method	Sep 2024	Oct–Dec 2024 (transition)	2025-S1	2025-S2	2026-S1
Static split (PIT)	93.5 (1.03) / 799	95.3 (0.45) / 1203	89.4 (1.04) / 455	94.0 (0.45) / 750	92.5 (0.58) / 535
Sliding window 60d	93.0 (1.04) / 742	92.0 (0.73) / 834	85.4 (1.16) / 323	92.9 (0.51) / 650	89.5 (0.76) / 419
ACI (γ=0.02)	93.0 (1.06) / 759	93.5 (0.55) / 915	85.7 (1.16) / 345	93.5 (0.49) / 669	89.1 (0.76) / 405
ACI (γ=0.05)	92.4 (1.16) / 718	92.1 (0.67) / 822	86.6 (1.14) / $405^{2.8}$	92.5 (0.58) / $609^{11.8}$	89.8 (0.75) / 446
Transport+ACI (γ=0.02)	92.6 (1.07) / 713	91.1 (0.8) / 787	86.5 (1.17) / $437^{0.6}$	92.4 (0.6) / $986^{1.1}$	90.3 (0.73) / 459
Transport+ACI (γ=0.05)	92.0 (1.19) / 681	90.9 (0.81) / 806	87.4 (1.13) / $446^{9.9}$	91.6 (0.63) / $718^{10.7}$	90.5 (0.72) / 489
Base-model CRPS (MWh)	96.1	133.5	70.5	94.5	79.0

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Table G.17: Effect of bounding the online update of α from below, whole test period. A bound of $\alpha_{\min} = 0.005$ removes infinite intervals entirely at negligible cost in coverage, and makes the unrestricted interval score finite and therefore comparable across methods.

Bound on α_t	Coverage (se)	Width	% infinite	IS finite	IS all
ACI (γ=0.05)					
unbounded	90.1 (0.43)	544	4.2	753	∞
$\alpha_{\min} = 0.005$	89.8 (0.43)	550	0.0	753	753
$\alpha_{\min} = 0.01$	89.8 (0.43)	536	0.0	739	739
$\alpha_{\min} = 0.02$	89.7 (0.43)	525	0.0	729	729
Transport+ACI (γ=0.05)					
unbounded	90.1 (0.43)	596	5.9	813	∞
$\alpha_{\min} = 0.005$	89.6 (0.43)	641	0.0	850	850
$\alpha_{\min} = 0.01$	89.6 (0.43)	598	0.0	807	807
$\alpha_{\min} = 0.02$	89.5 (0.43)	559	0.0	770	770

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Table G.18: Sensitivity to alternative definitions of the transition window, hurdle base model. W_1 is the Wasserstein-1 distance between the positive log-magnitudes of the window and those of the calibration set. Width and interval score are means over finite intervals, with the fraction of infinite intervals beside them.

Method	Coverage (se)	Width	% inf.	IS finite
Oct–Dec 2024 (the one used), W_1 = 1.253				
Static split	95.3 (0.45)	1203	0.0	1395
ACI (γ=0.02)	93.5 (0.55)	915	0.0	1158
Transport+ACI (γ=0.05)	90.9 (0.81)	806	0.0	1107
Sep–Dec 2024, W_1 = 1.179				
Static split	94.9 (0.43)	1105	0.0	1279
ACI (γ=0.02)	93.4 (0.49)	877	0.0	1091
Transport+ACI (γ=0.05)	91.1 (0.68)	775	0.0	1038
Nov 2024–Jan 2025, W_1 = 1.109				
Static split	94.8 (0.79)	1105	0.0	1256
ACI (γ=0.02)	91.2 (1.17)	788	0.0	1019
Transport+ACI (γ=0.05)	88.6 (1.43)	795	5.5	1083
Oct 2024–Feb 2025, W_1 = 1.002				
Static split	94.5 (0.57)	1000	0.0	1164
ACI (γ=0.02)	91.2 (0.83)	732	0.0	965
Transport+ACI (γ=0.05)	89.5 (1.01)	787	8.1	1086
Oct–Nov 2024, W_1 = 1.342				
Static split	94.0 (0.58)	1108	0.0	1373
ACI (γ=0.02)	92.4 (0.73)	896	0.0	1219
Transport+ACI (γ=0.05)	89.9 (1.08)	842	0.0	1223
Sep 2024–Mar 2025, W_1 = 0.992				
Static split	94.3 (0.5)	968	0.0	1124
ACI (γ=0.02)	91.5 (0.72)	736	0.0	951
Transport+ACI (γ=0.05)	89.9 (0.87)	769	6.8	1040

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Table G.19: Sensitivity of the adaptive schemes to the step γ and to the length of the recent window, whole test period, September 2024 to May 2026. This table is a display produced after the selection of Section 4.7 was closed, never a criterion: the selection used only the inner validation set. Pure ACI has no window. Bold marks the configuration selected by rolling origin; the 60-day cells at $\gamma = 0.02$ and 0.05 are the configurations of Algorithm 1 reported in the main text. †The configuration produces infinite limits, so its unrestricted interval score is infinite.

	ACI	Transport+ACI, recent window (days)				
γ		30	45	60	90	120
Coverage (%)						
0.005	**90.9**	89.9	90.0	90.0	90.0	**90.0**
0.01	90.5	90.0	90.0	90.0	90.0	90.0
0.02	90.2	90.0	90.1	90.1	90.0	90.0
0.05	90.1	90.1	90.1	90.1	90.0	90.0
0.10	90.1	90.0	90.0	90.0	90.0	90.0
0.20	90.0	89.9	90.0	90.0	90.0	90.0
Infinite intervals (%)						
0.005	**0.0**	0.0	0.0	0.0	0.0	**0.0**
0.01	0.0	0.0	0.0	0.0	0.0	0.0
0.02	0.0	0.0	0.3	0.5	0.6	1.1
0.05	4.2	5.3	6.4	5.9	5.2	4.4
0.10	1.6	6.2	4.2	4.4	3.6	4.3
0.20	15.1	22.5	23.0	21.9	22.4	22.9
Interval score over finite intervals (MWh)						
0.005	**763**	775	759	753	756	**761**
0.01	748	803	789	776	769	770
0.02	743	898	905†	871†	846†	822†
0.05	753†	826†	806†	813†	831†	895†
0.10	784†	893†	804†	821†	817†	784†
0.20	936†	872†	877†	769†	774†	769†

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Table G.20: Calibration schemes designed to address panel dependence, whole test period. The last four columns give the worst absolute deviation from the 90% nominal level over the groups of each axis, in percentage points, which is the measure of conditional coverage; smaller is better. Width and interval score are means over finite intervals, with the fraction of infinite intervals beside them.

	Marginal				Worst deviation from nominal (pp)			
Scheme	Coverage (se)	Width	% inf.	IS finite	Technology	Size	Region	High-curt.
Pooled split	92.5 (0.37)	678	0.0	826	2.5	3.6	4.8	2.9
Block conformal, one plant per date	92.7 (0.36)	697	0.0	840	2.8	3.9	5.1	3.1
Mondrian by technology	92.3 (0.37)	651	0.0	809	4.1	3.7	5.1	2.7
Mondrian by region	92.1 (0.38)	596	0.0	764	4.0	3.7	7.2	2.5
Mondrian by capacity tercile	92.5 (0.37)	674	0.0	823	2.6	3.9	4.8	2.9
Plant-level conformal	92.5 (0.39)	636	0.0	773	4.1	3.0	7.2	2.8
ACI (γ=0.02), for reference	90.2 (0.43)	551	0.0	743	0.3	1.5	4.7	1.7
Transport+ACI (γ=0.05), for reference	90.1 (0.43)	596	5.9	813	0.3	1.6	4.8	3.3

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